

The adductor canal

❑ Also termed as (subsartorial canal, Hunter canal)

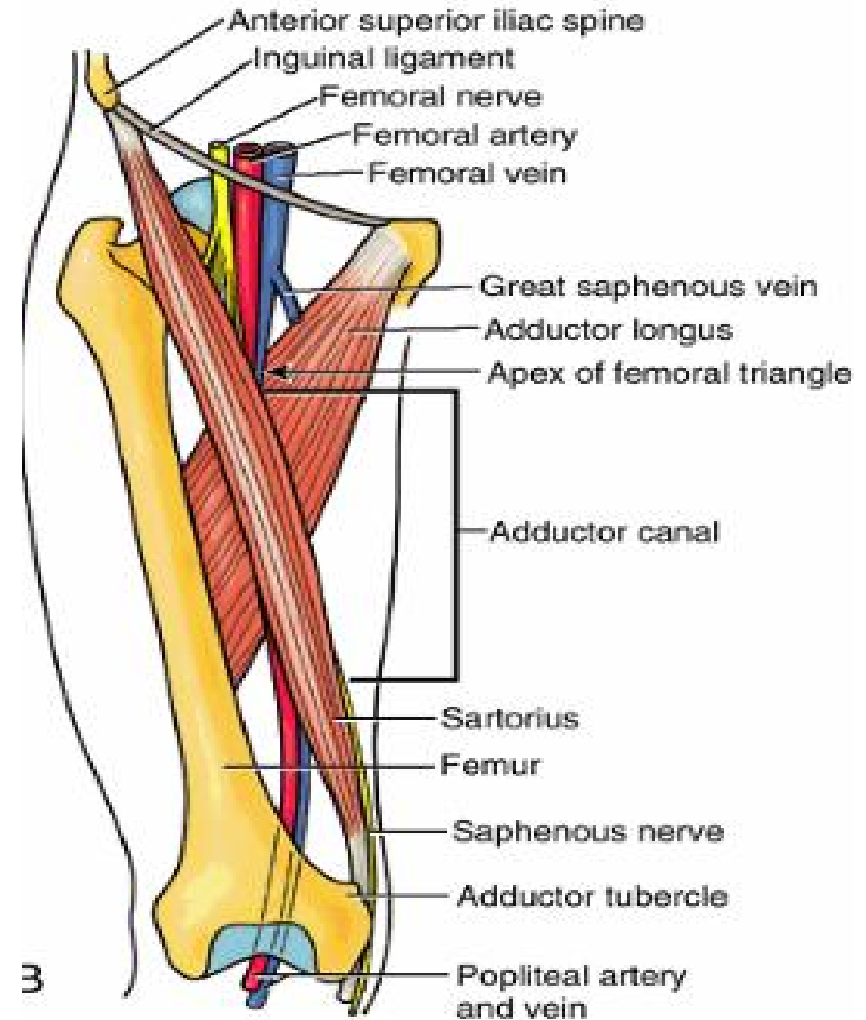
❑ Extends from the apex of the femoral triangle

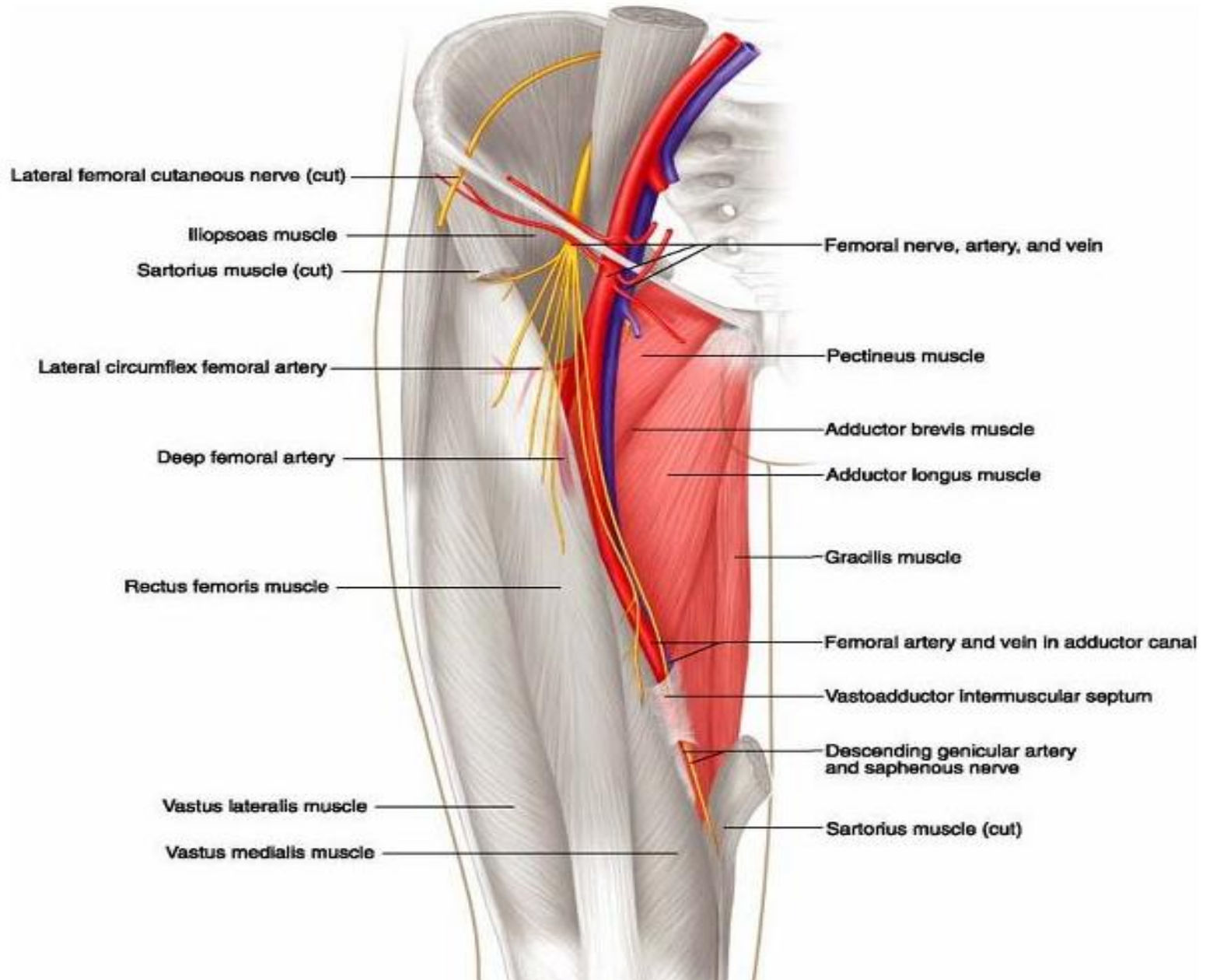
❑ BOUNDARY

- **anteriorly** and **laterally** by the vastus medialis
- **posteriorly** by the adductor longus and adductor magnus
- **medially** by the sartorius

❑ Contents

- **Femoral artery and vein,**
- **Saphenous nerve, and**
- **Nerve to vastus medialis**

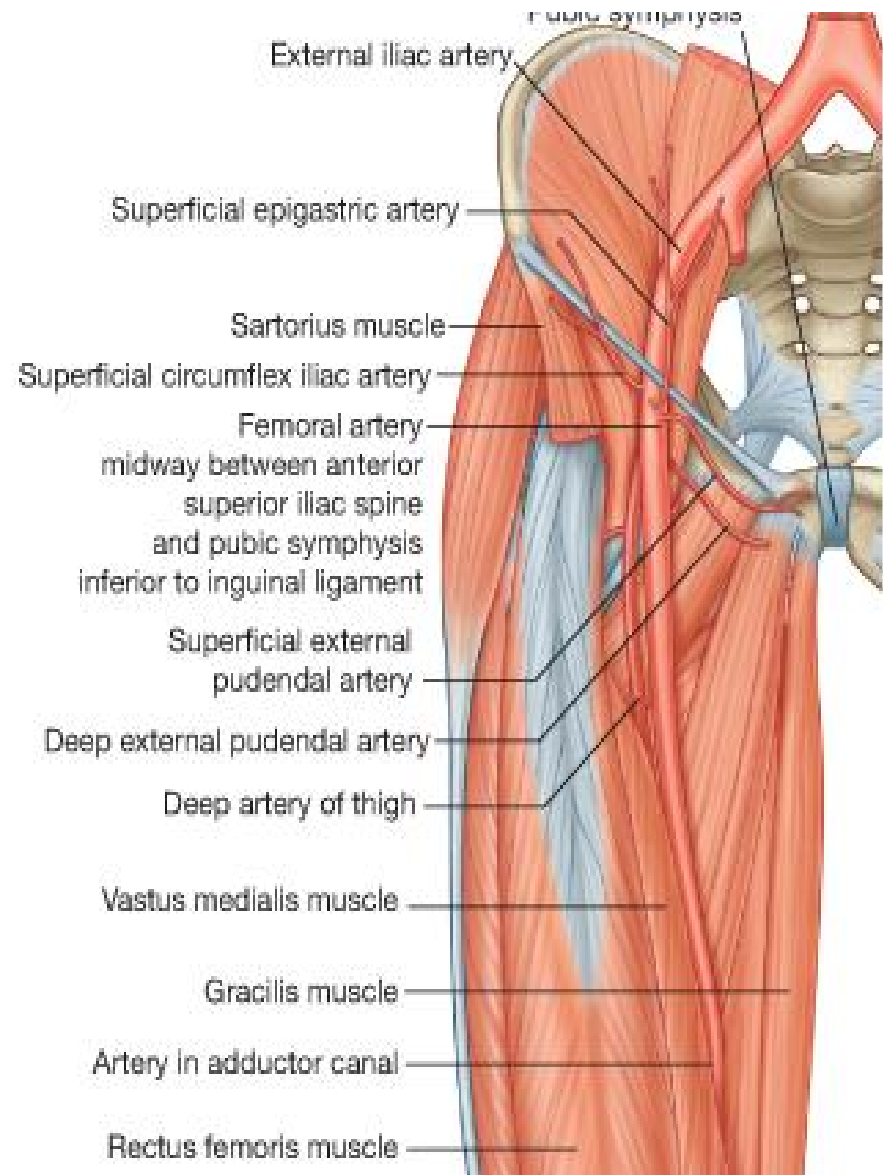


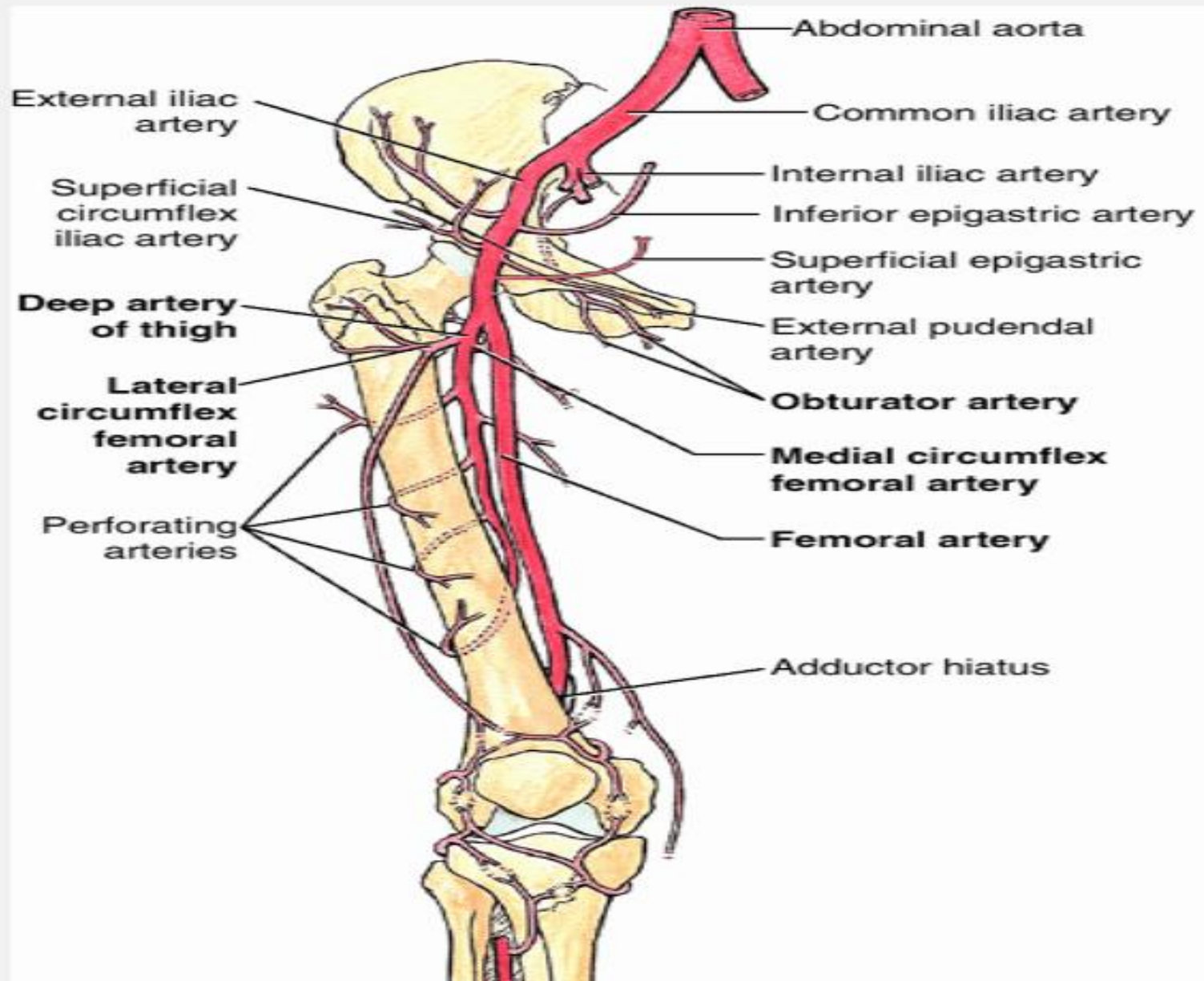


Neurovasculature in Anteromedial Thigh

The femoral artery

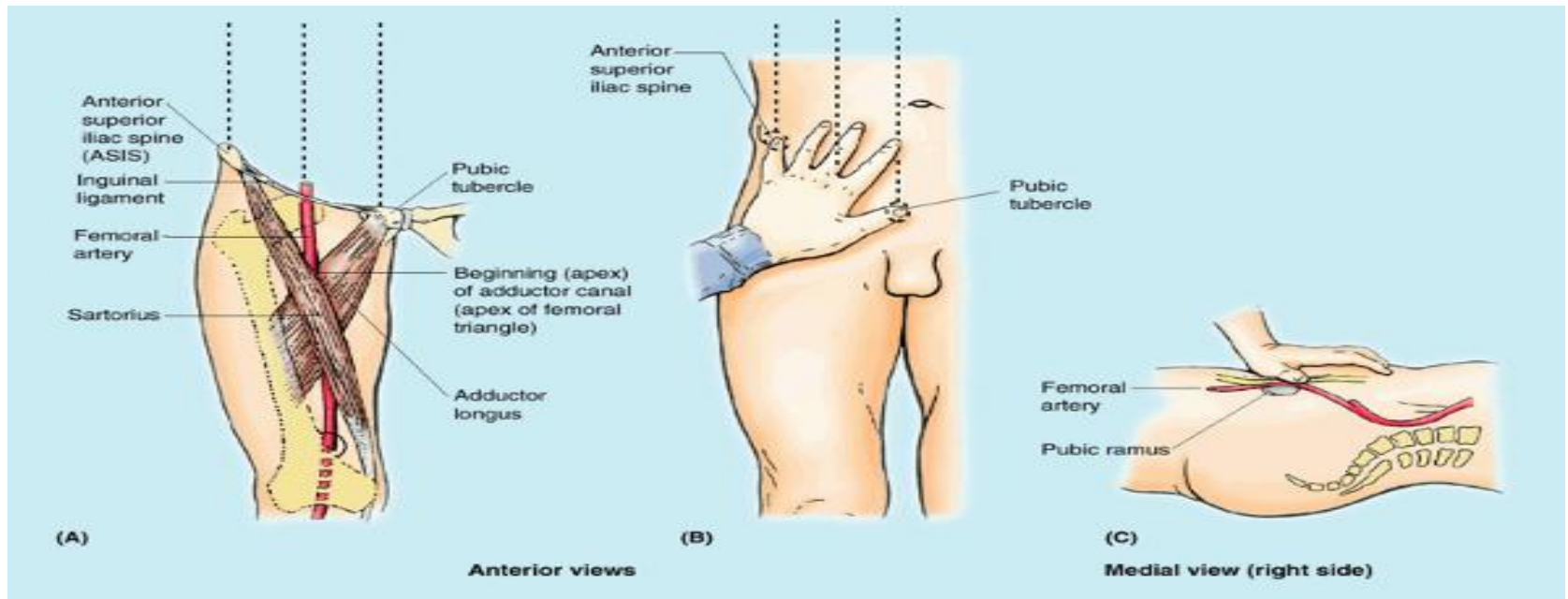
- The chief aa to the lower limb
- Branch of external iliac aa
- Branches
 - Superficial & deep external pudendal aa
 - supply scrotum and labium major
 - Superficial epigastric artery
 - Superficial circumflex iliac artery
 - **Deep femoral** (profunda femoris) artery :- Medial & lateral circumflex femoral and perforating branch)





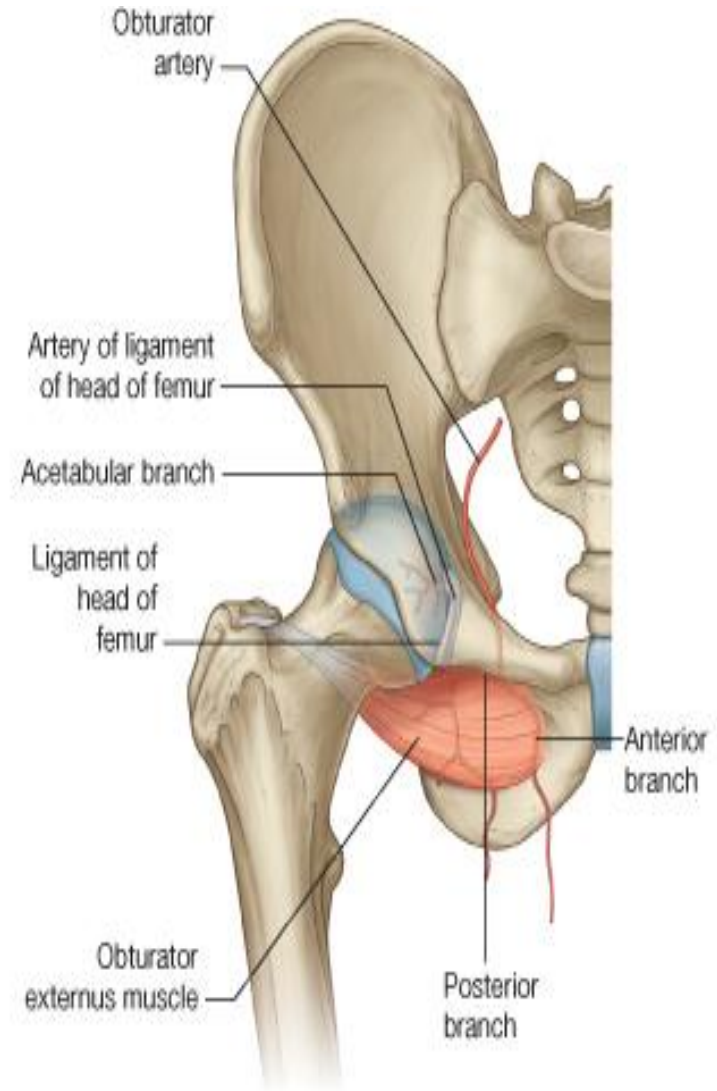
Femoral pulse

- Palpable just inferior to the midpoint of the inguinal ligament
- The femoral artery may be compressed here to control arterial bleeding after lower limb trauma
- The femoral artery may be cannulated just inferior to the midpoint of the inguinal ligament (e.g., for cardioangiography, radiography of the heart and great vessels after the introduction of contrast material)



Obturator artery

- Arises from **internal iliac artery**
- Passes through obturator foramen
- Enters medial compartment of the thigh
- Divides into anterior and posterior branches
- Supplies **obturator externus and adductors of thigh**
- Posterior branch gives off acetabular branch to head of femur

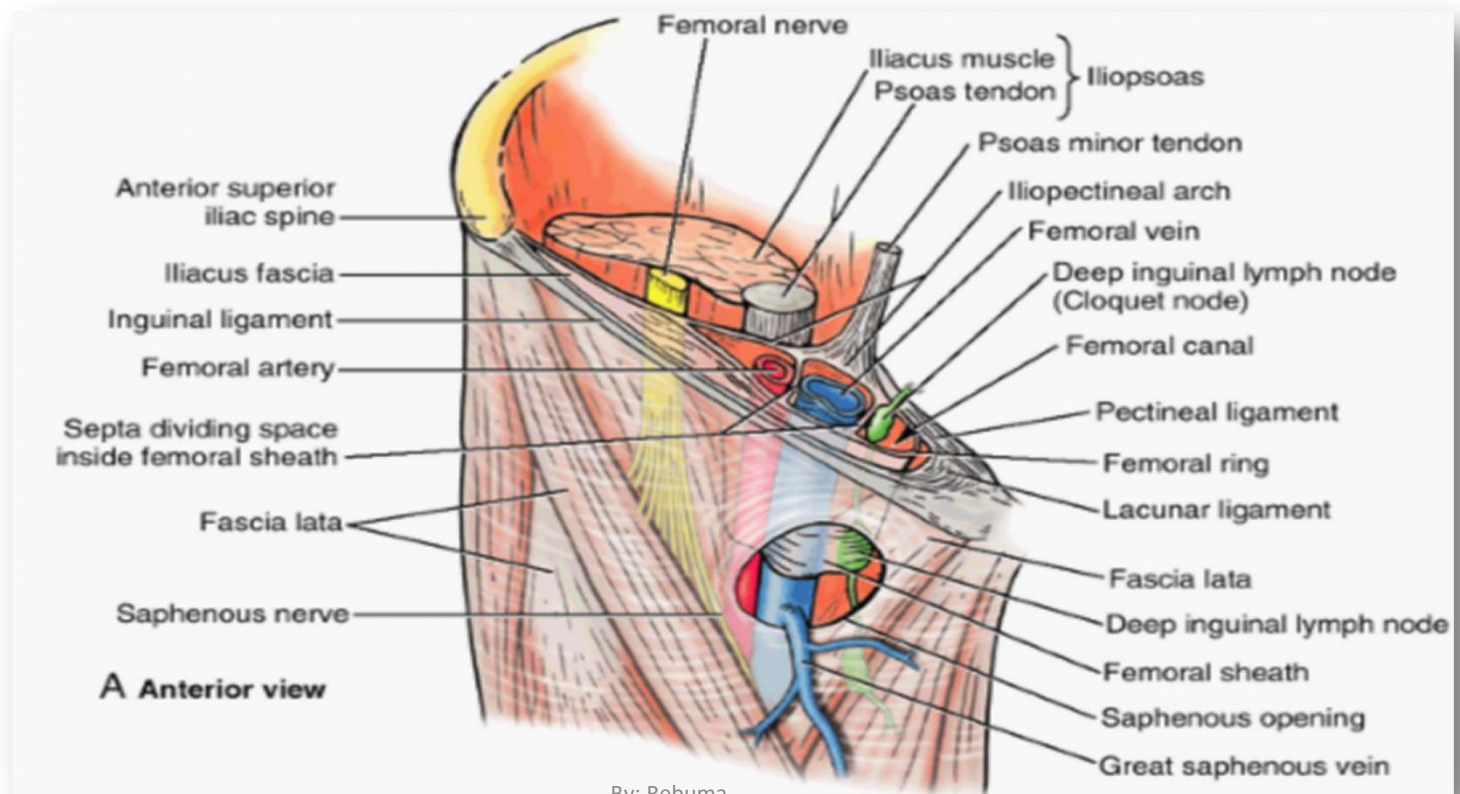


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The femoral vein

- Is the continuation of the popliteal vein at adductor hiatus
- Tributaries
 - Descending genicular
 - Profunda femoris
 - circumflex femoral
 - Deep external pudendal
 - Great saphenous

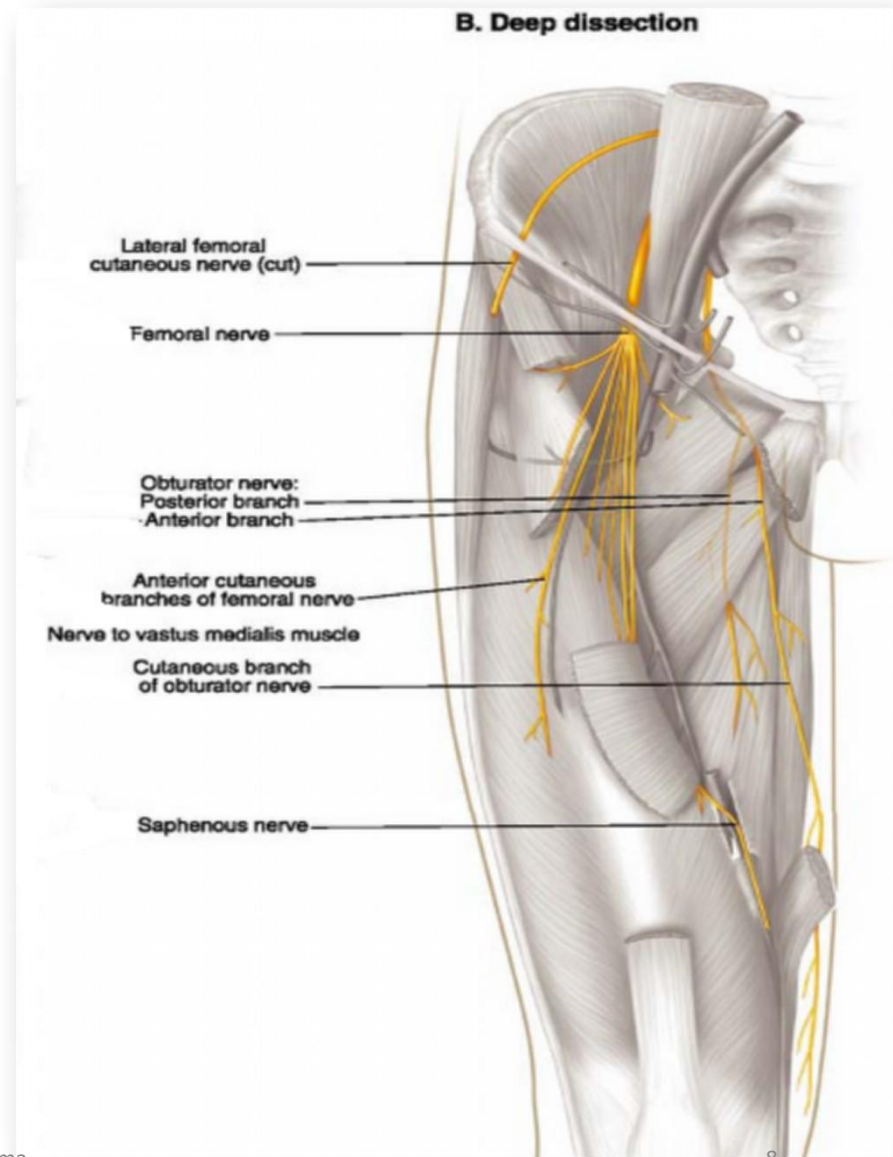
Ends at external iliac vein



NERVES

Femoral Nerve:- L2- L4

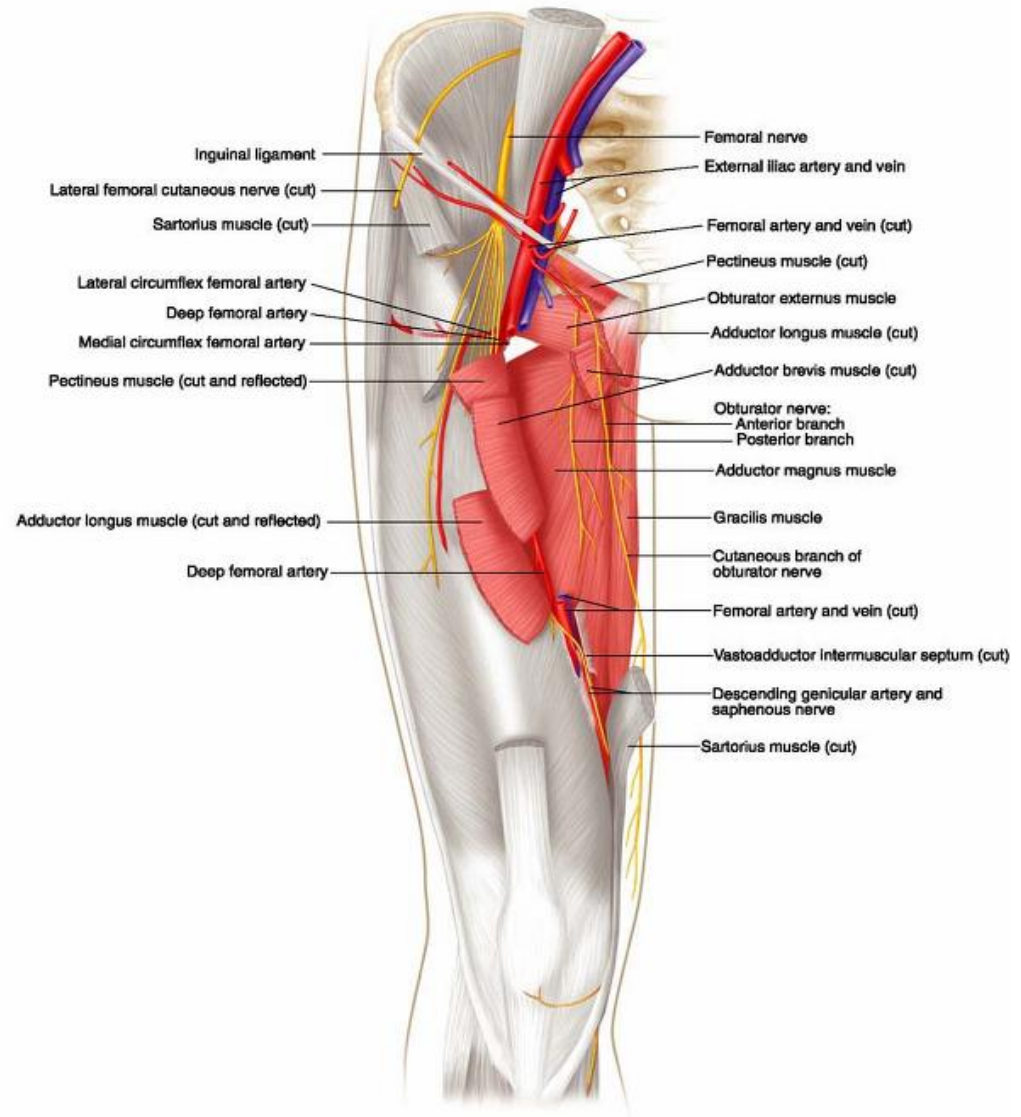
- is the largest branch of the lumbar plexus.
- divides into several terminal branches to the anterior thigh muscles
- sends articular branches to the hip and knee joints and
- provides cutaneous branches to the anteromedial thigh
- Terminal cutaneous branch - **saphenous** nerve



NERVE

The obturator nerve L2-L4

- It divides into anterior and posterior branches,
- **The anterior branch** supplies the adductor longus, adductor brevis, gracilis, and pectineus;
- **The posterior branch** supplies the obturator externus and adductor magnus



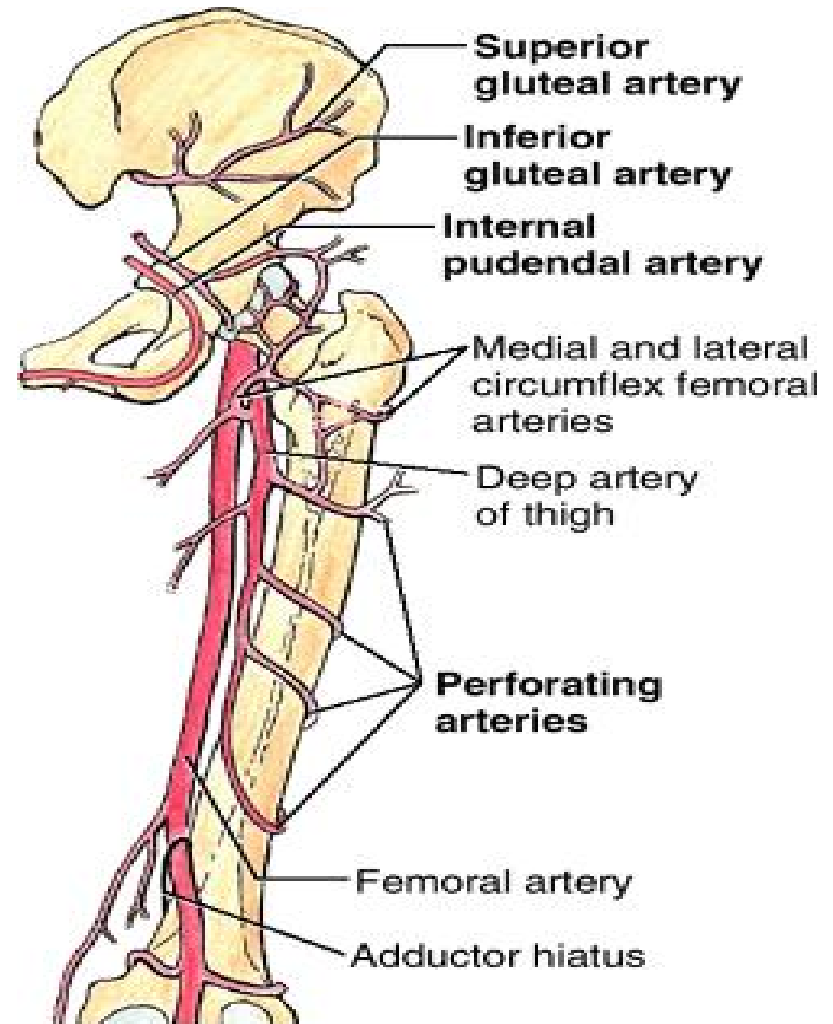
Vasculature of Gluteal Region and Posterior Thigh

Arteries

- ❑ Superior gluteal aa
- ❑ Inferior gluteal aa
- ❑ Internal pudendal- **supply perineum**
 - ❖ all arises from internal iliac aa
- ❖ The deep artery of the thigh - giving off perforating arteries to posterior compartment of the thigh

❖ Veins

- ✓ Accompany the arteries
- ✓ Have same name with arteries
- ✓ Are tributaries of the internal iliac veins



Posterior view

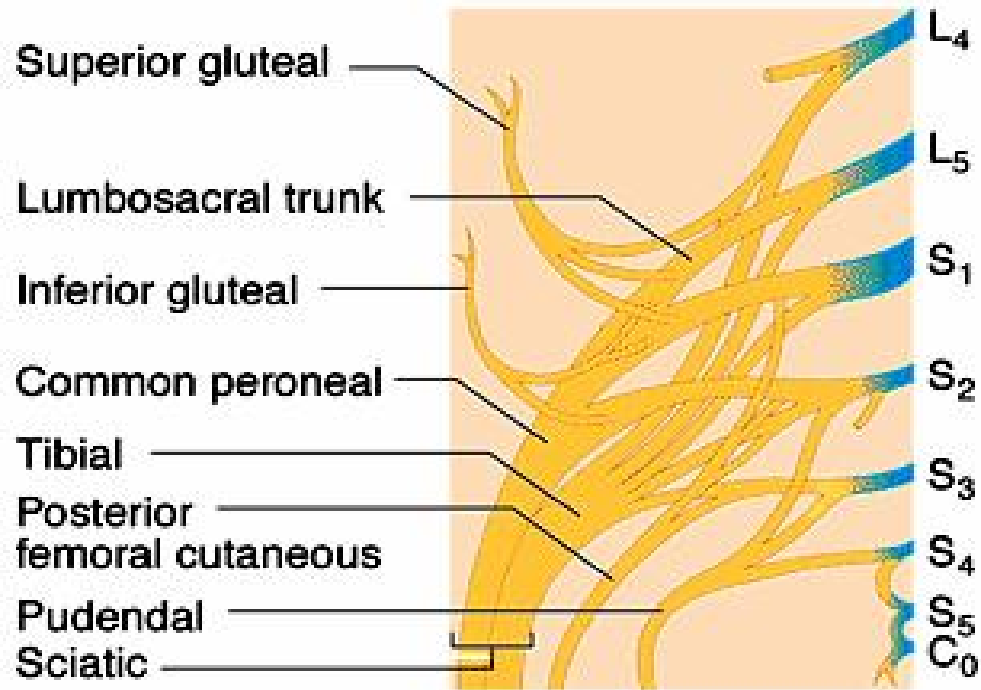
Nerves of Gluteal Region and Posterior Thigh

➤ Several nerves arise from the **sacral plexus** (L4, L5, S1–S4) and supply the gluteal region and posterior thigh

➤ Branches

- Superior gluteal nerve - gluteus medius, minimus, tensor fascia lata
- Inferior gluteal nerve – gluteus maximus
- Nerve to quadratus femoris
- Nerve to piriformis
- Nerve to obturator internus
- Posterior cutaneous nerve of the thigh
- Pudendal nerve
- Sciatic nerve

Sacral Plexus

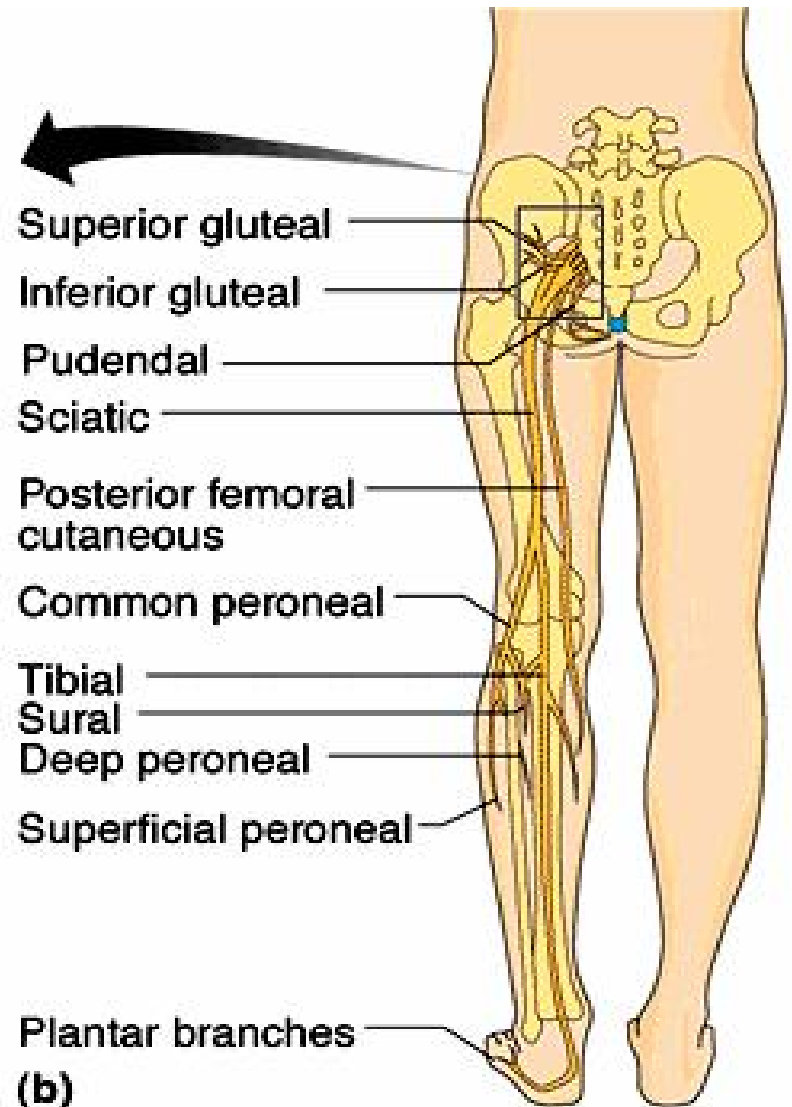


(a)

Key:

■ = Roots

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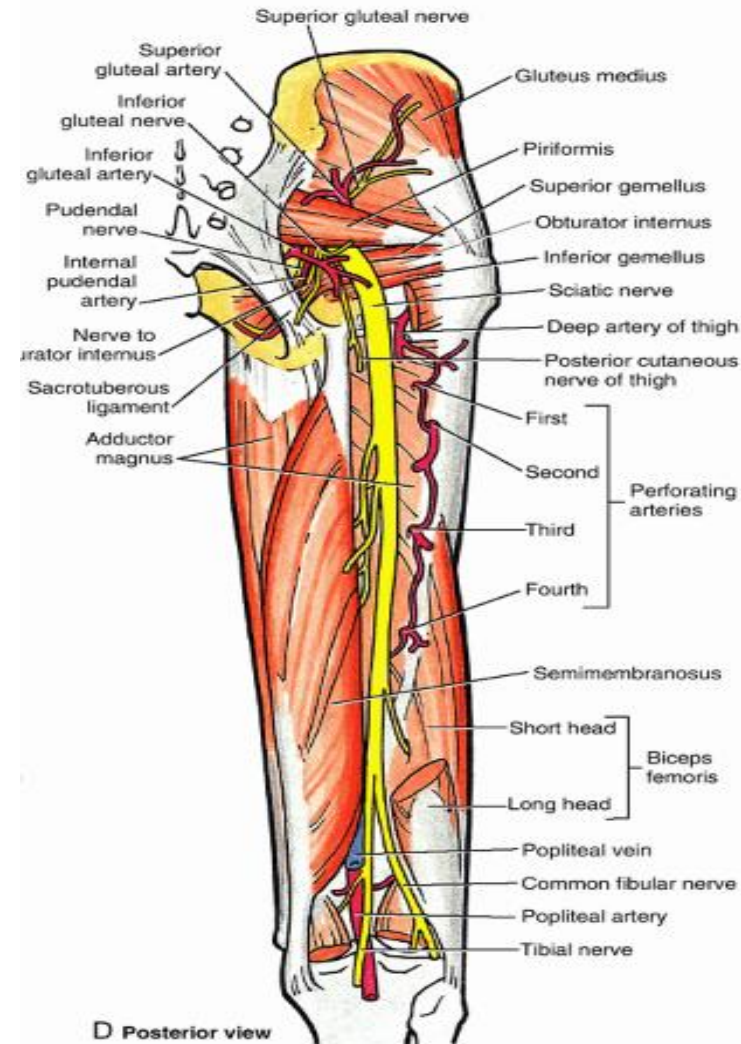


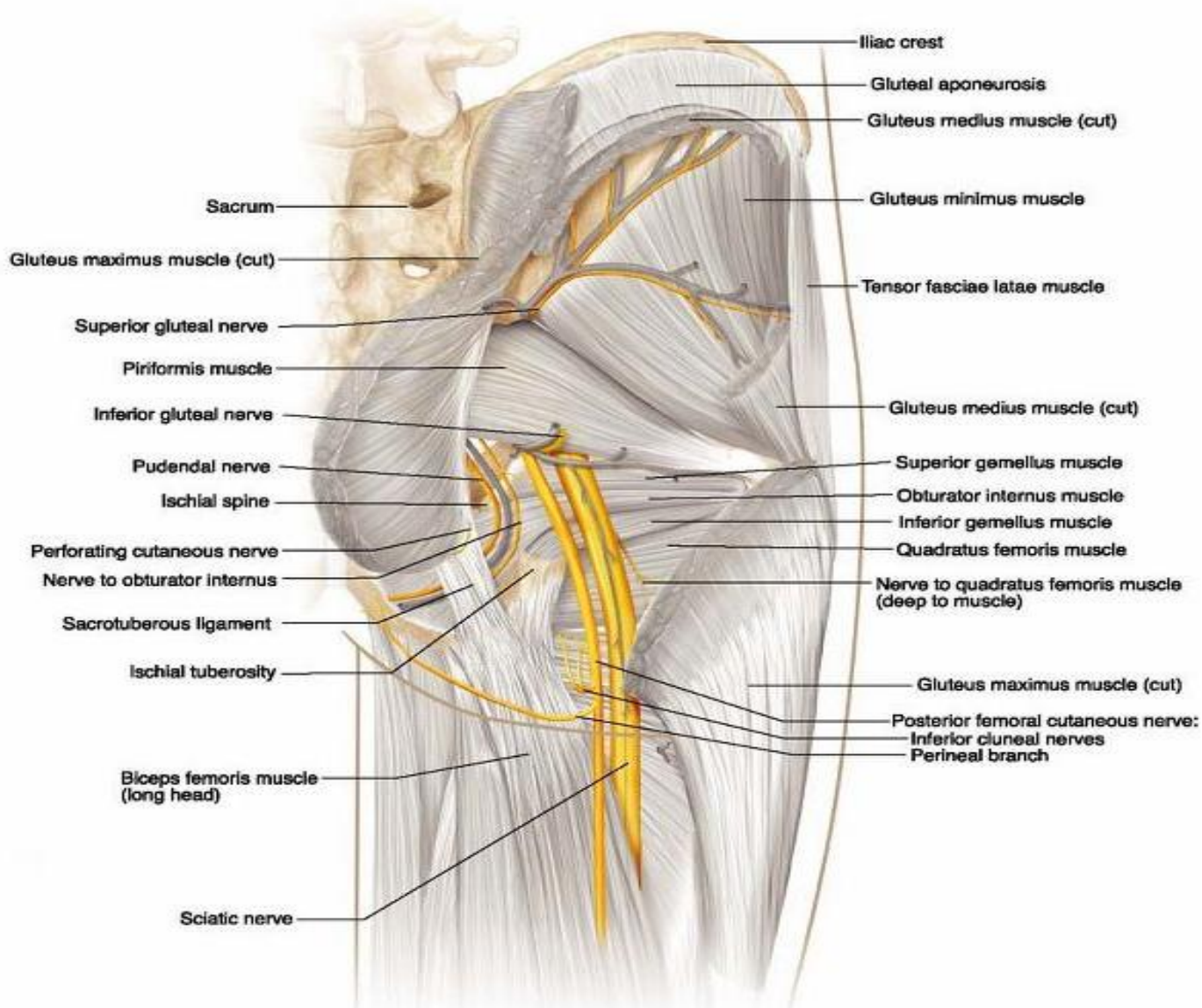
(b)

Nerves of Gluteal Region and Posterior Thigh

- **Sciatic nerve** (L4-S3)

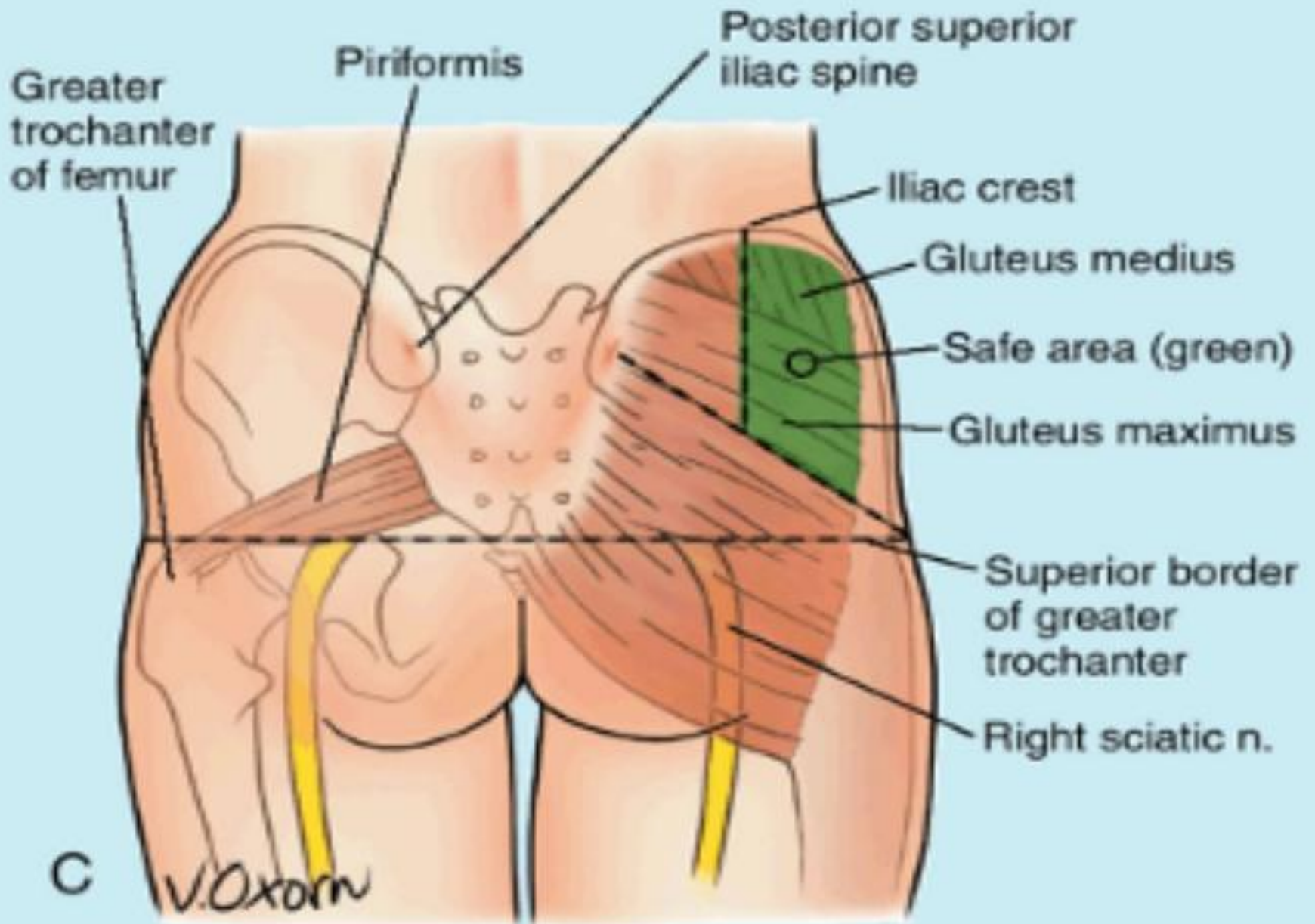
- the largest nerve in the body
- the continuation of the main part of the **sacral plexus**
- is really two nerves loosely bound together in the same connective tissue sheath: **the tibial nerve, and the common fibular nerve**
- Terminate in the middle of the thigh by dividing into tibial and common fibular nerves
- Supplies **no structures** in the gluteal region
- It supplies the posterior thigh muscles, all leg and foot muscles, and the skin of posterior thigh, most of the leg and the foot

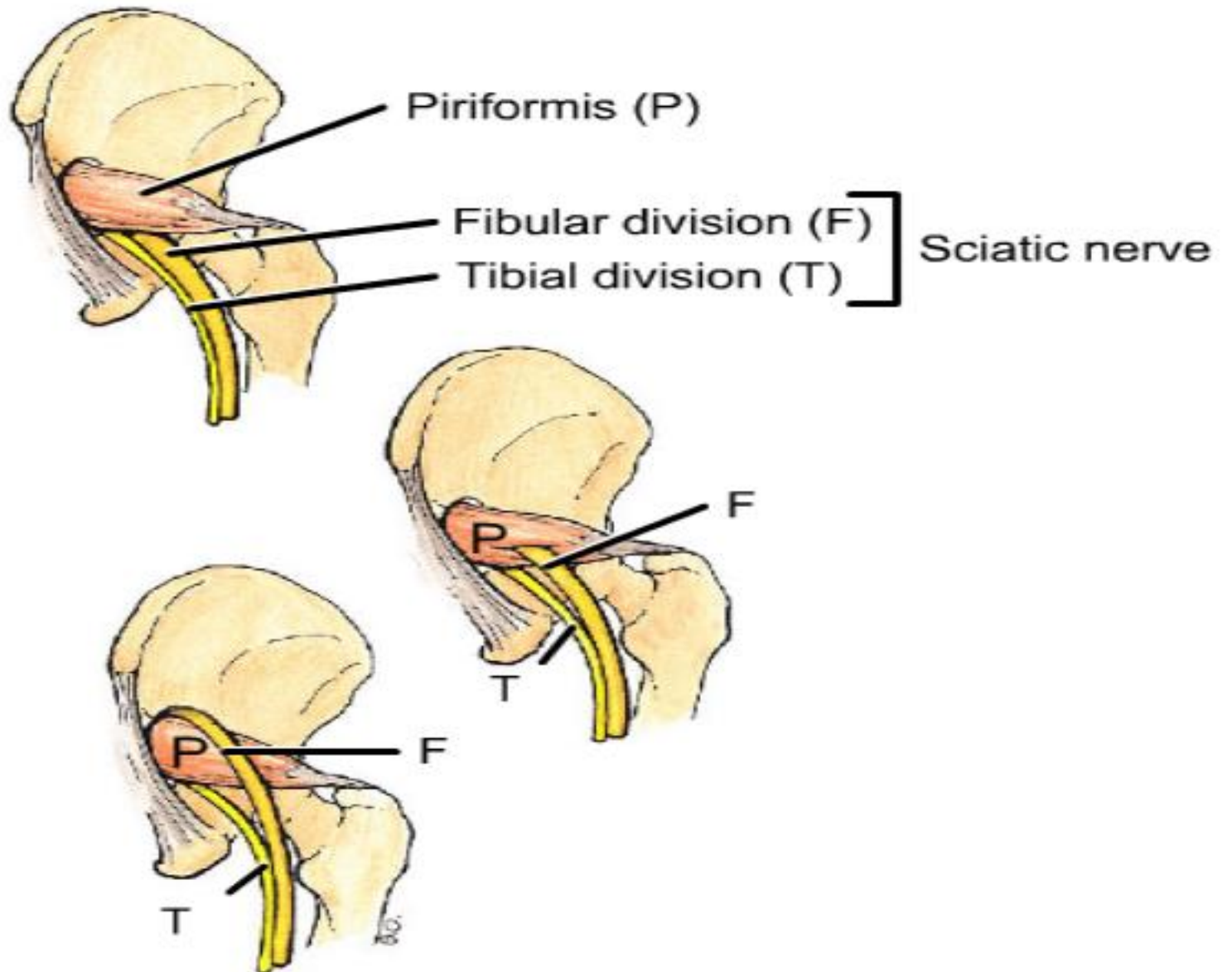




Intra-gluteal injections

- The gluteal region is a common site for intramuscular injection of drugs
- The gluteal region is a **favorable injection** site because the muscles are thick and large; consequently, they provide a substantial **volume for absorption of injected substances by intramuscular veins**
- Complications of improper technique include nerve injury, hematoma, and abscess formation
- Injections into the buttock are safe only in the **superolateral** quadrant of the buttock





THE POPLITEAL FOSSA

- Is a mostly fat-filled **diamond-shaped** space posterior to the knee

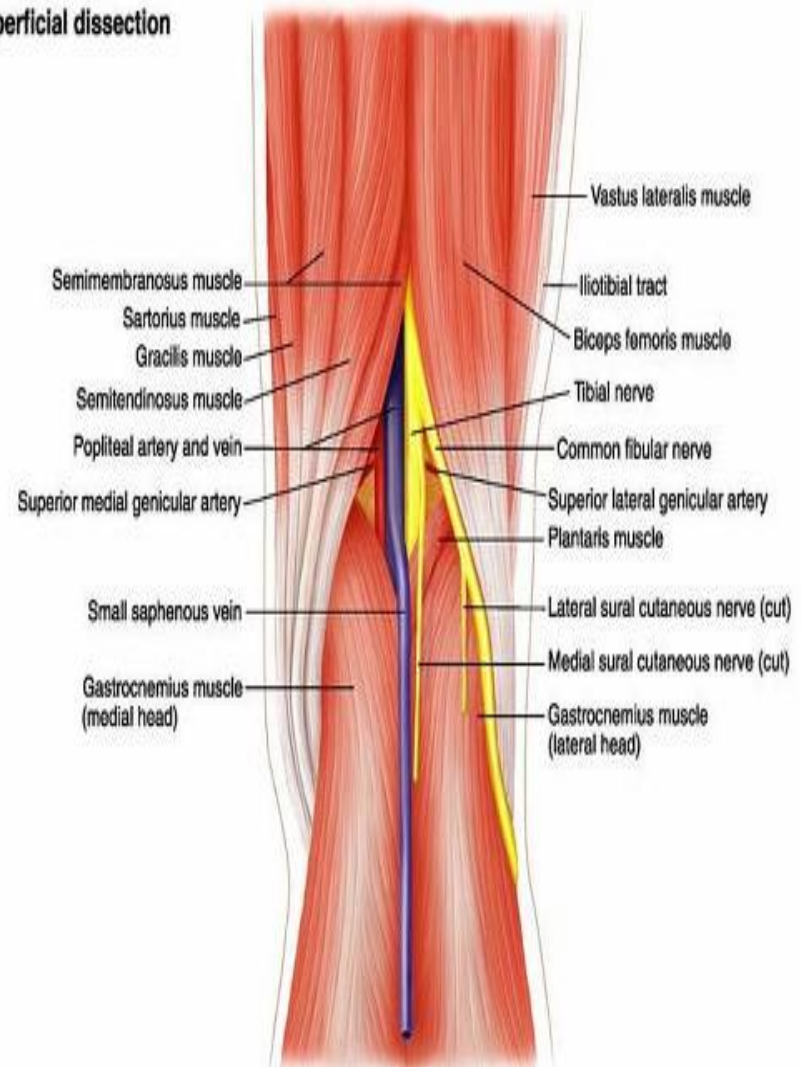
The popliteal fossa is **bound**

- **Superolateral** - biceps femoris
- **Superomedially**: semimembranosus.
- **Inferolateral** and **inferomedially** by the lateral and medial heads of the gastrocnemius,

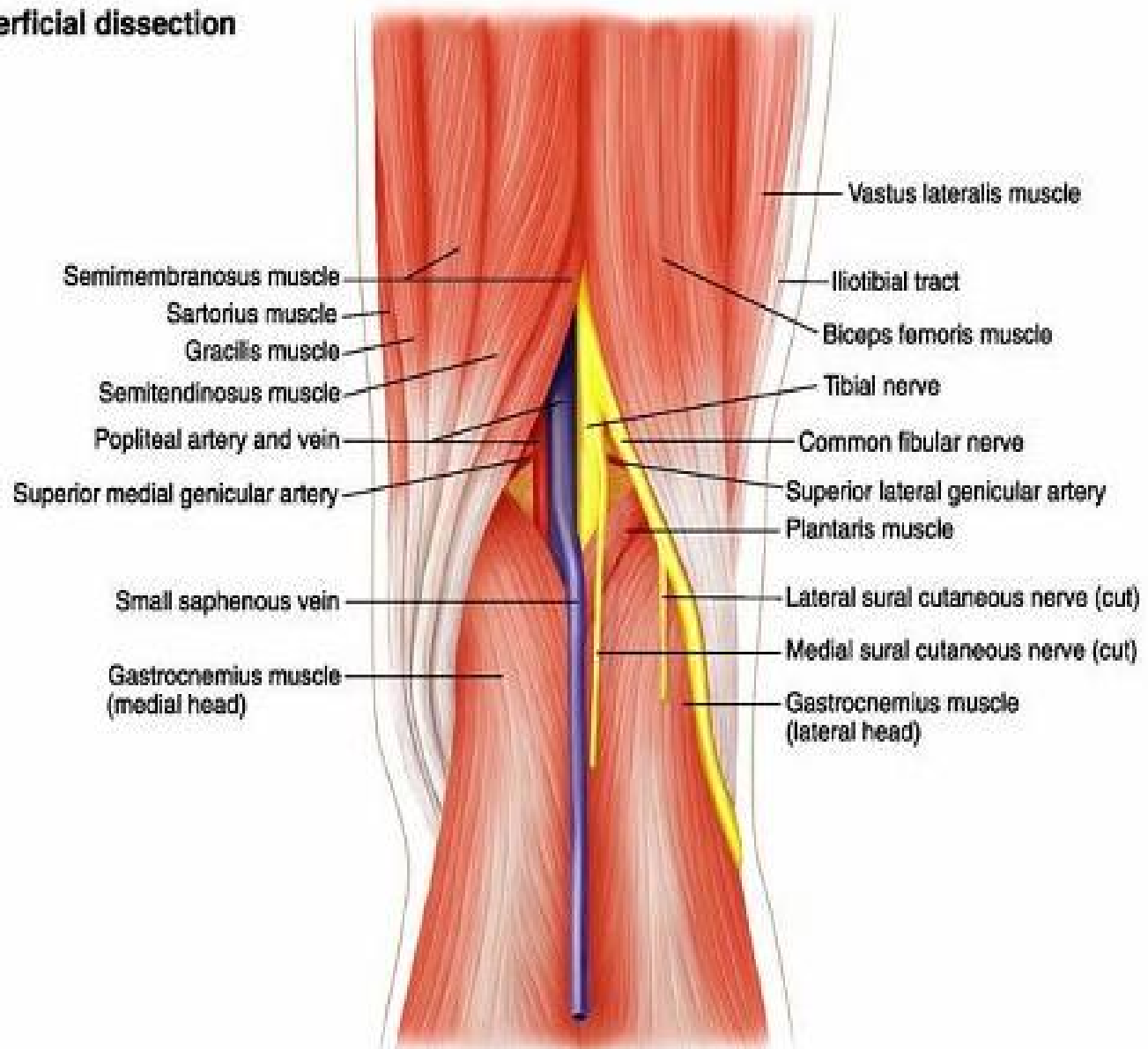
ROOF: Posteriorly by skin and popliteal fascia

FLOOR: **Anteriorly** by the popliteal surface of the femur, posterior capsule of the knee joint, and the popliteus fascia covering the popliteus muscle .

A. Superficial dissection



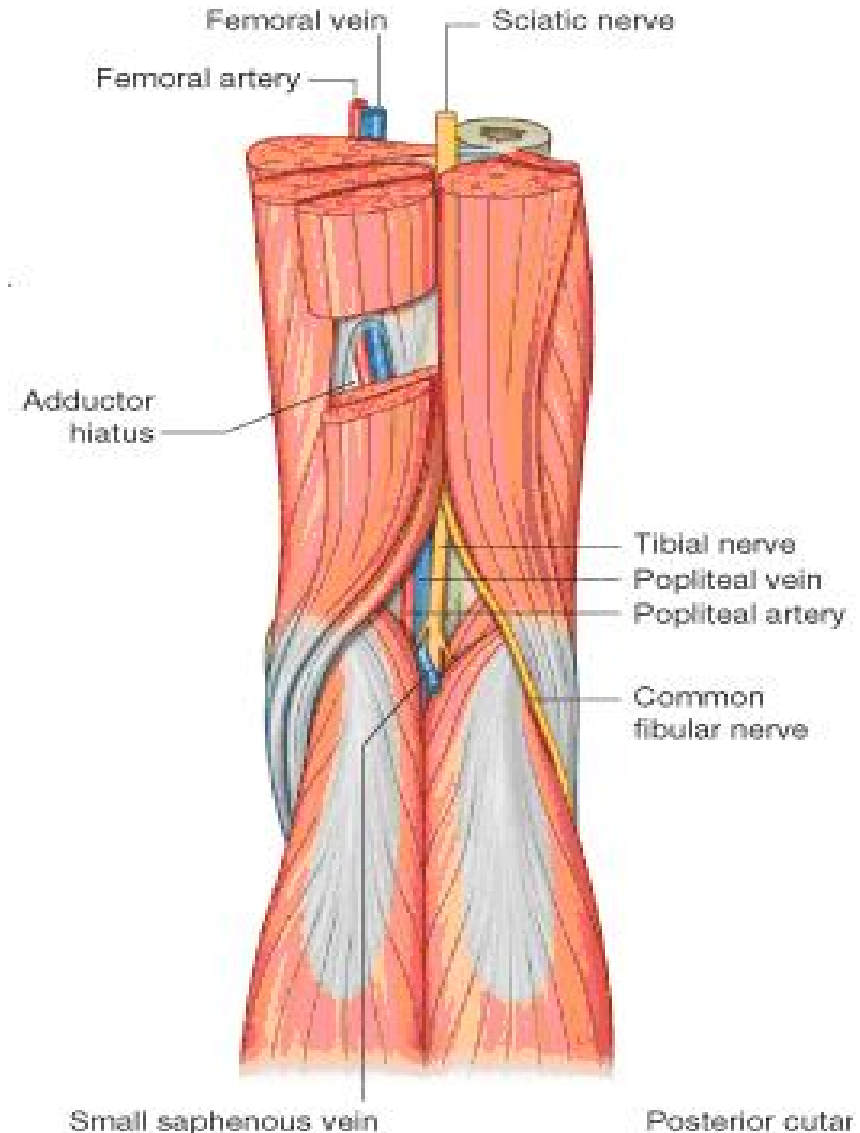
A. Superficial dissection



THE POPLITEAL FOSSA

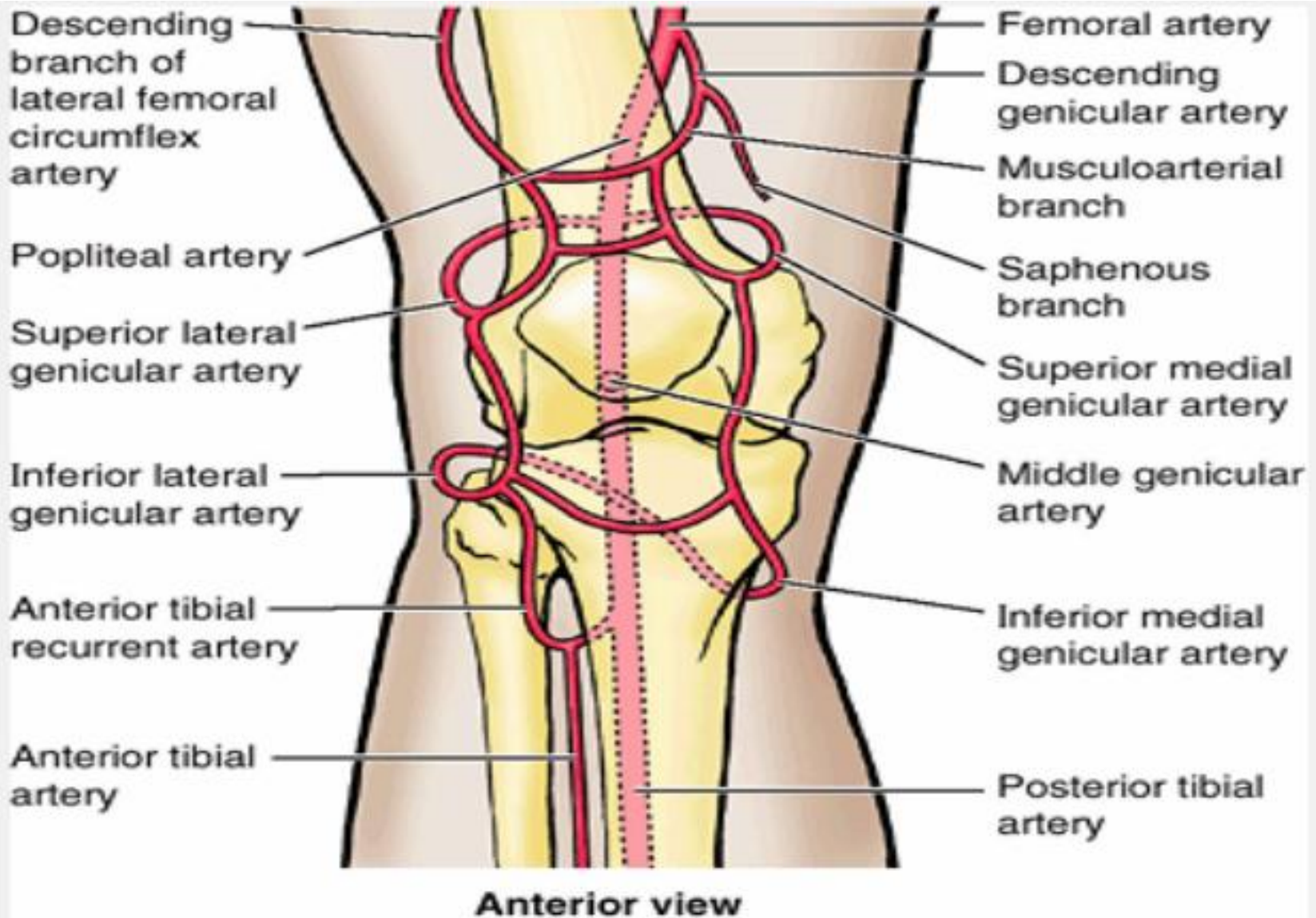
The contents

- Termination of the small saphenous vein.
- Popliteal artery and vein and their branches and tributaries.
- Tibial and common fibular nerves
- Posterior cutaneous nerve of the thigh.
- Popliteal lymph nodes and lymphatic vessels.



Popliteal artery

- Continuation of the femoral artery
- At the inferior border of the popliteus it divides into **anterior** and **posterior tibial** arteries
- The deepest structure in the popliteal fossa
- Branches
 - Five genicular branches and of the popliteal artery supply the
 - superior lateral
 - superior medial
 - Middle
 - inferior lateral
 - inferior medial
 - Branch for muscles, joint capsule and ligaments of the knee joint
 - Anterior and posterior tibial arteries



Popliteal vein

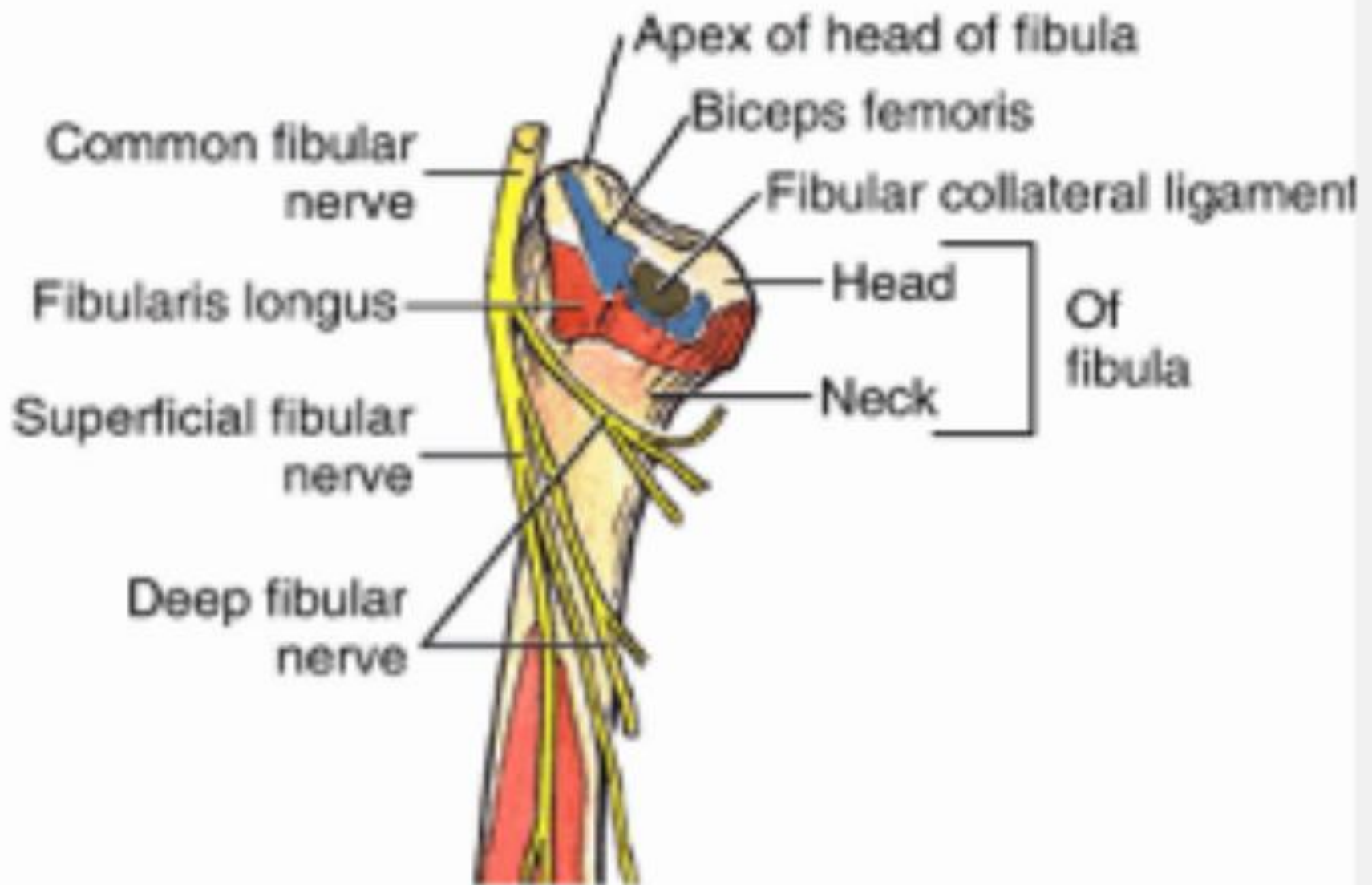
- Begins as an assembly of venae comitantes below the knee
- Formed at distal end of popliteus
- Terminate in the adductor hiatus
- Tributaries
 - main – vena comitantes of anterior and posterior tibial artery
 - muscular and cutaneous
 - genicular veins
 - small saphenous vein

Tibial nerve

- The medial, **larger** terminal branch of the sciatic nerve
- The most superficial of the three main central components of the popliteal fossa (nerve, vein, and artery)
- A medial sural cutaneous nerve is also derived from the tibial nerve joined by the sural communicating branch of the common fibular nerve to form the **sural nerve**
 - This nerve supplies the lateral side of the leg and ankle

Common fibular nerve

- The lateral, smaller terminal branch of the sciatic nerve
- Winds around the fibular neck, where it is vulnerable to injury
- Here it divides into its terminal branches, the **superficial and deep fibular nerves**



B Lateral view

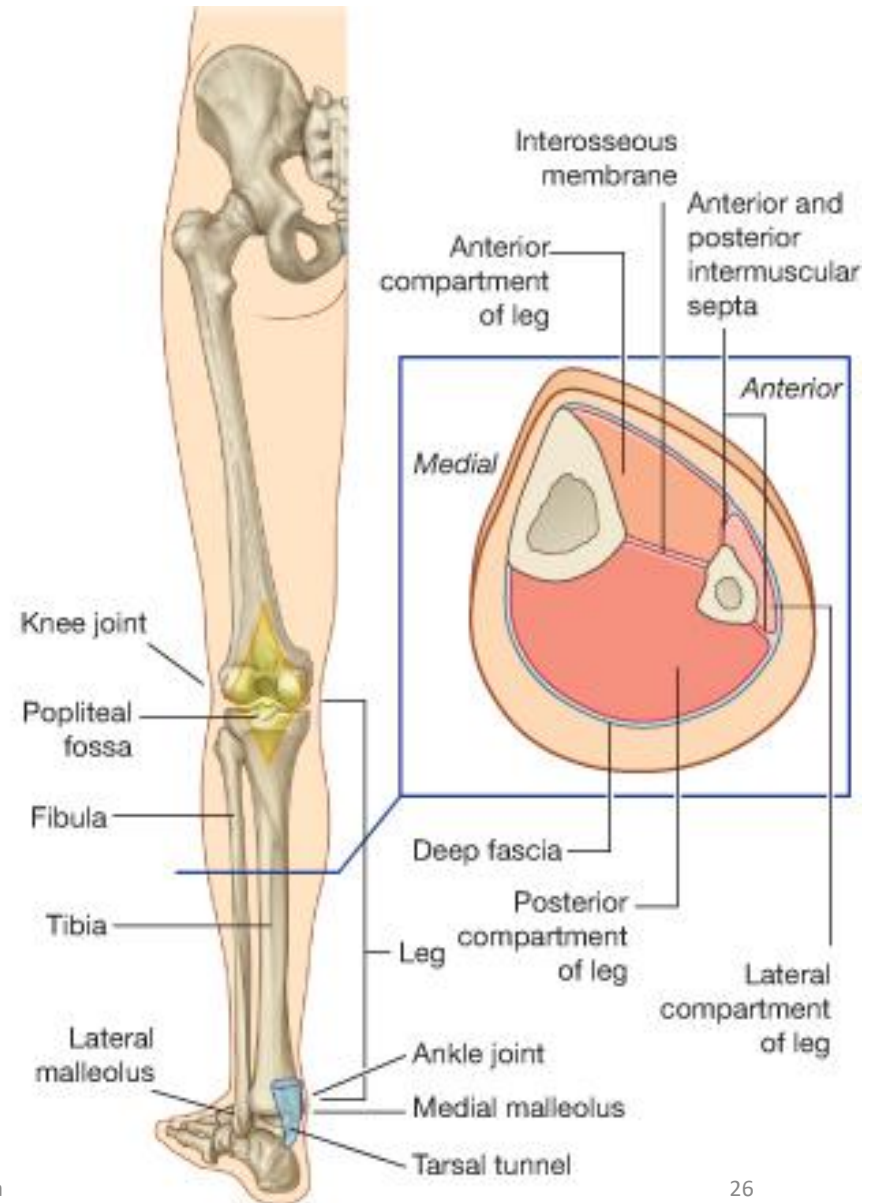
THE LEG

✓ The leg is divided into
three compartments by
intermuscular septa
interosseous membrane and
two leg bones

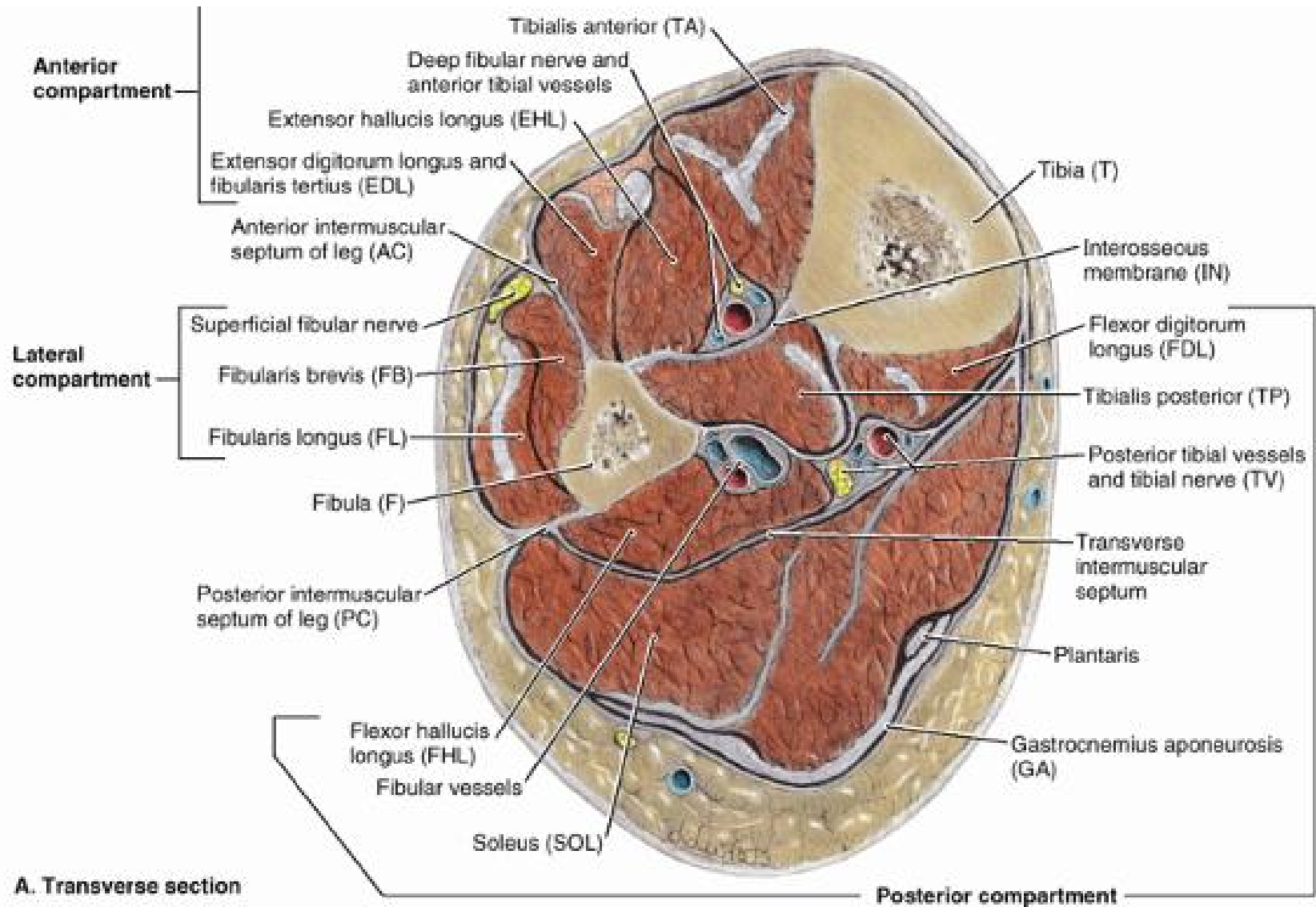
■ Anterior

■ Lateral

■ Posterior

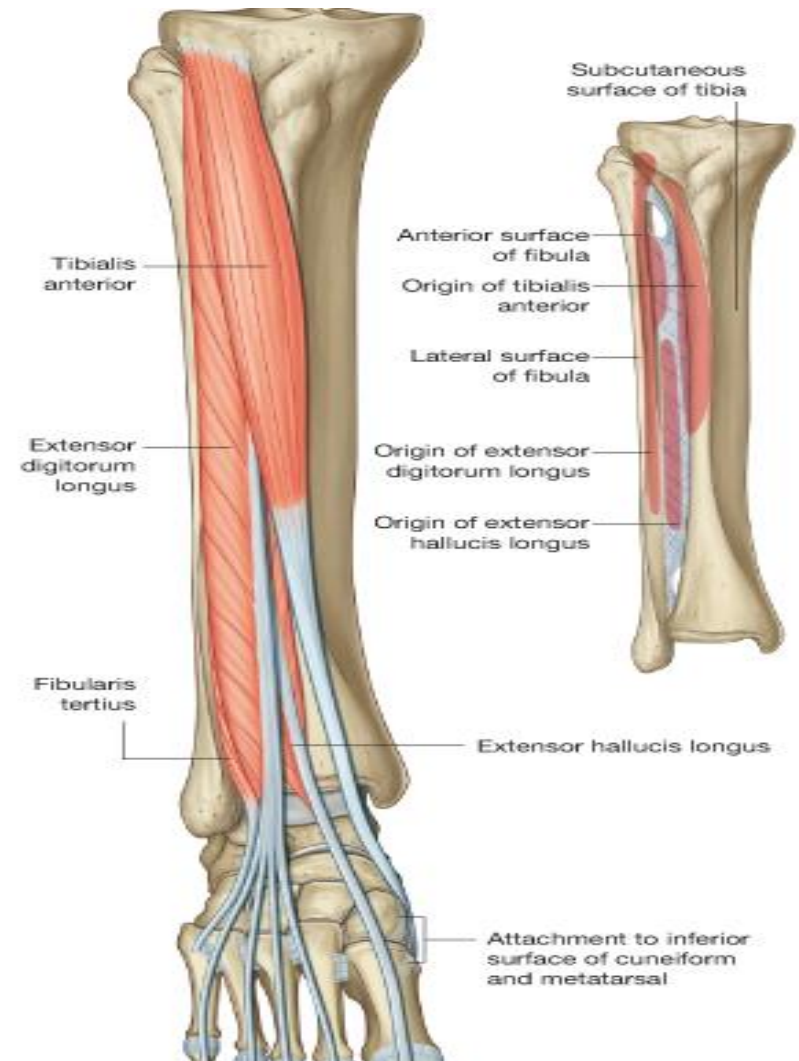


THE LEG



Anterior Compartment of Leg

- Are Dorsiflexor (extensor compartment) and anterior to the interosseous membrane
- Four muscles in the anterior compartment
 1. Tibialis anterior
 2. Extensor digitorum longus
 3. Extensor hallucis longus
 4. Fibularis tertius



Anterior Compartment of Leg

Muscle	Proximal attachment	Distal attachment	Innervation	action
Tibialis anterior	Lateral surface of tibia	medial cuneiform and base of metatarsal I	Deep fibular nerve	Dorsiflexion of foot at ankle joint; inversion of foot
Extensor hallucis longus	Middle one-half of medial surface of fibula	base of distal phalanx of great toe	Deep fibular nerve	Extension of great toe and dorsiflexion of foot
Extensor digitorum longus	Proximal one-half of medial surface of fibula and lateral tibial condyle	bases of distal and middle phalanges of lateral four toes	Deep fibular nerve	Extension of lateral four toes and dorsiflexion of foot
Fibularis tertius	Distal part of medial surface of fibula	base of metatarsal V	Deep fibular nerve	Dorsiflexion and eversion of foot

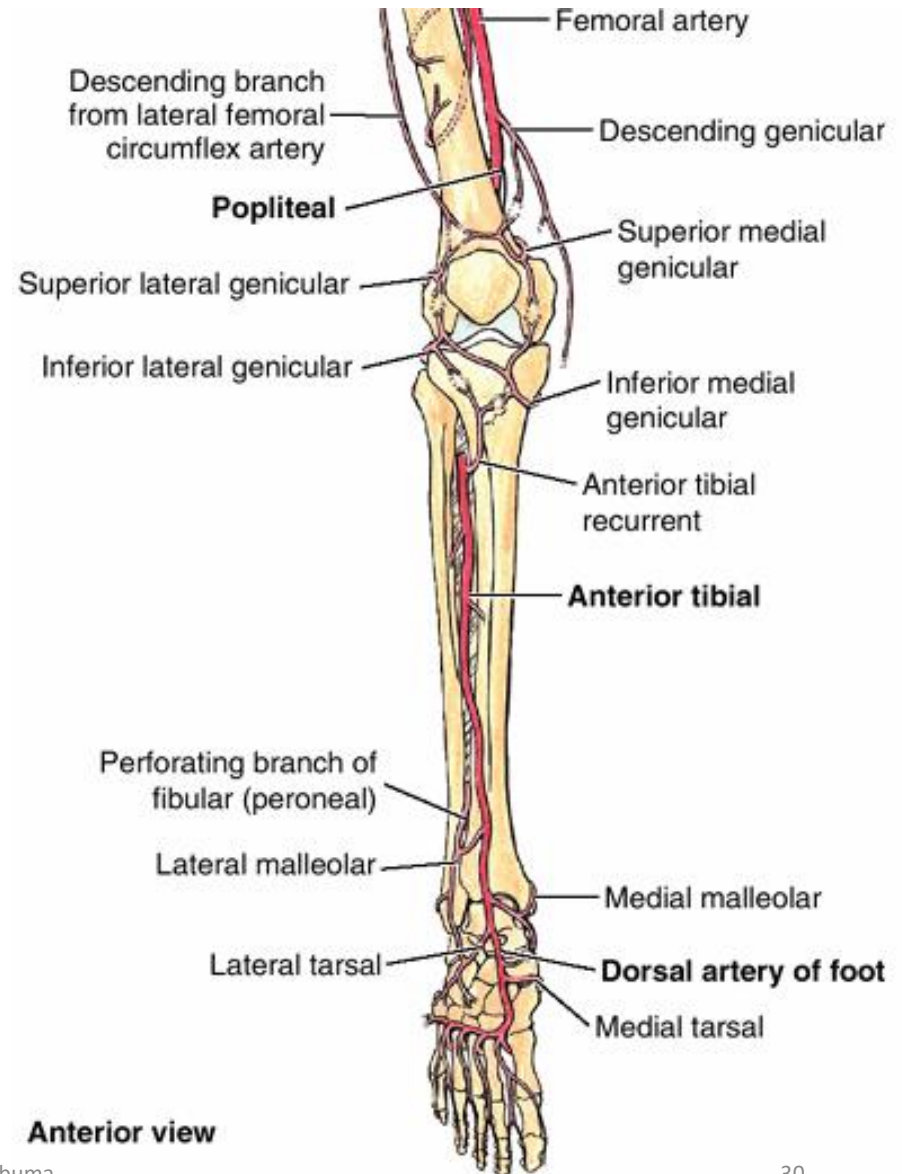
Neurovasculature in the leg

✓ Arteries

- The **anterior tibial artery** supplies structures in the anterior compartment
- It ends at the ankle joint, midway between the malleoli, where it becomes the dorsal artery of the foot

✓ Veins

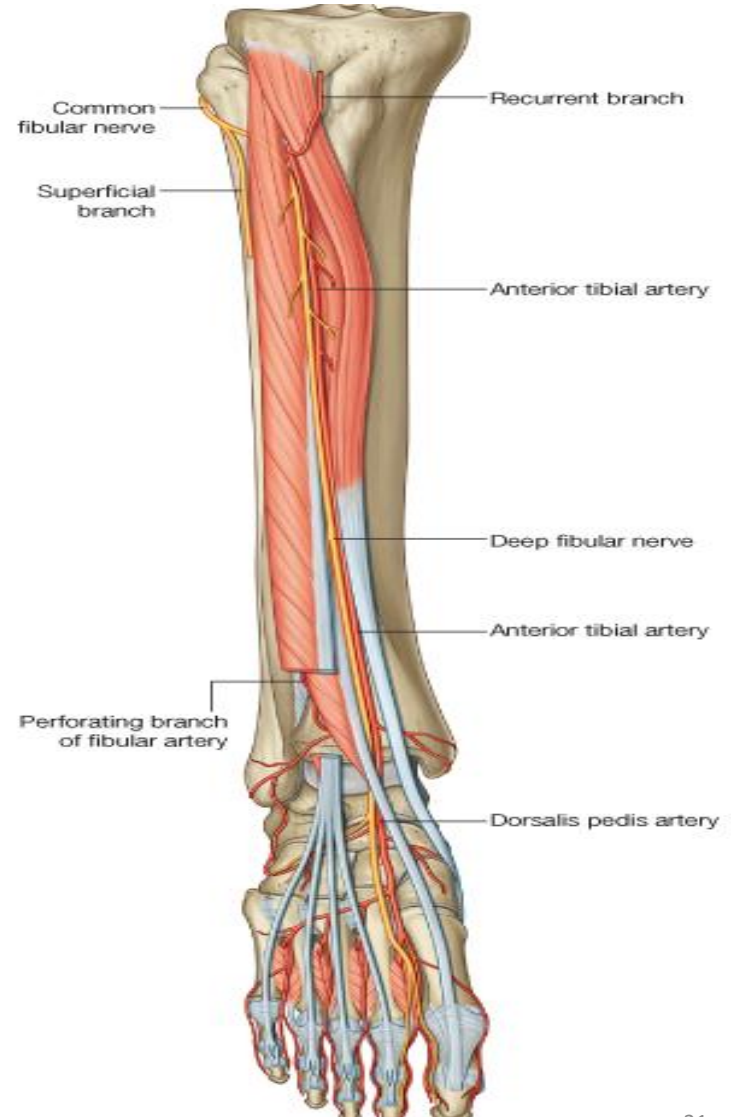
- Deep veins follow the arteries and have similar names.



Neurovasculature in the ant. Comp.

✓ Nerves

- The **deep fibular nerve**, one of the two terminal branches of the common fibular nerve, is the nerve of the anterior compartment.



Lateral compartment of Leg

The lateral compartment

Is **evertor** compartment,

contains **two muscles**:

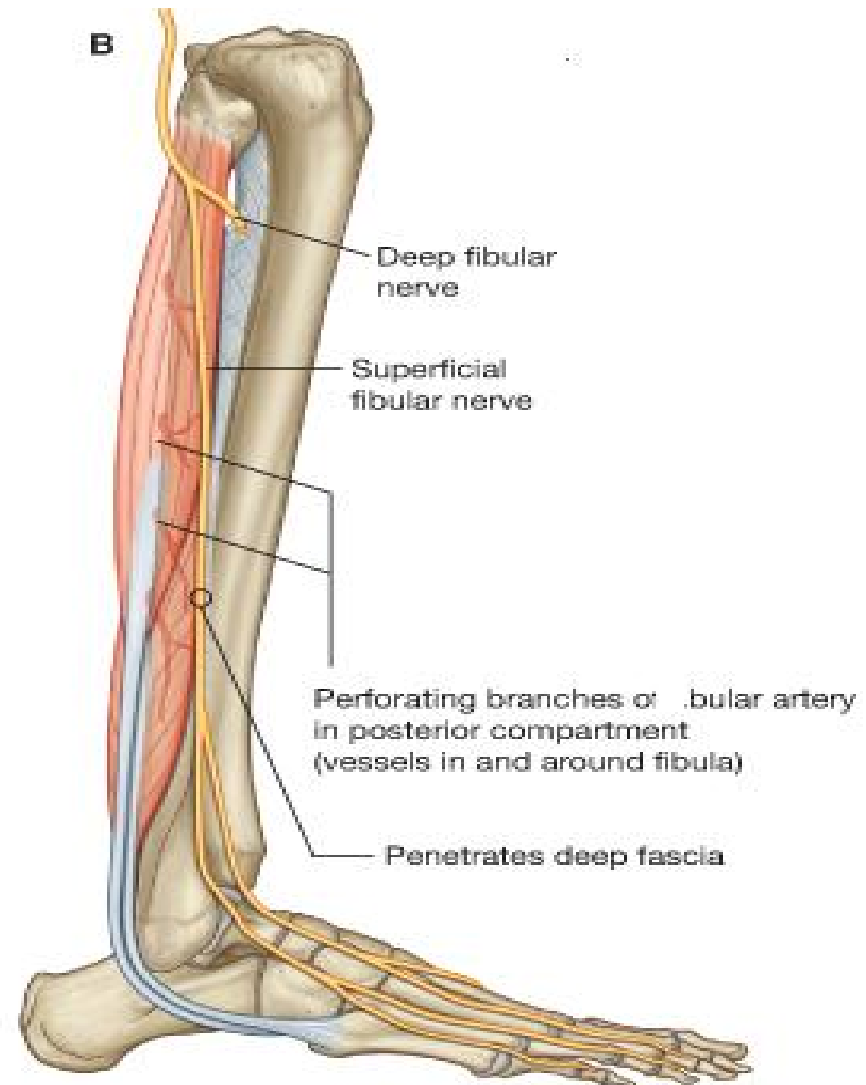
1. The fibularis longus
and
2. The fibularis brevis

Lateral compartment of Leg

Muscle	Proximal attachment	Distal attachment	Innervation	action
Fibularis longus	Upper 2/3 of lateral surface of fibula	medial cuneiform and base of metatarsal I	Superficial fibular nerve	Eversion and plantarflexion of foot
Fibularis brevis	Lower two-thirds of lateral surface of fibula	Lateral tubercle at base of metatarsal V	Superficial fibular nerve	Eversion of foot

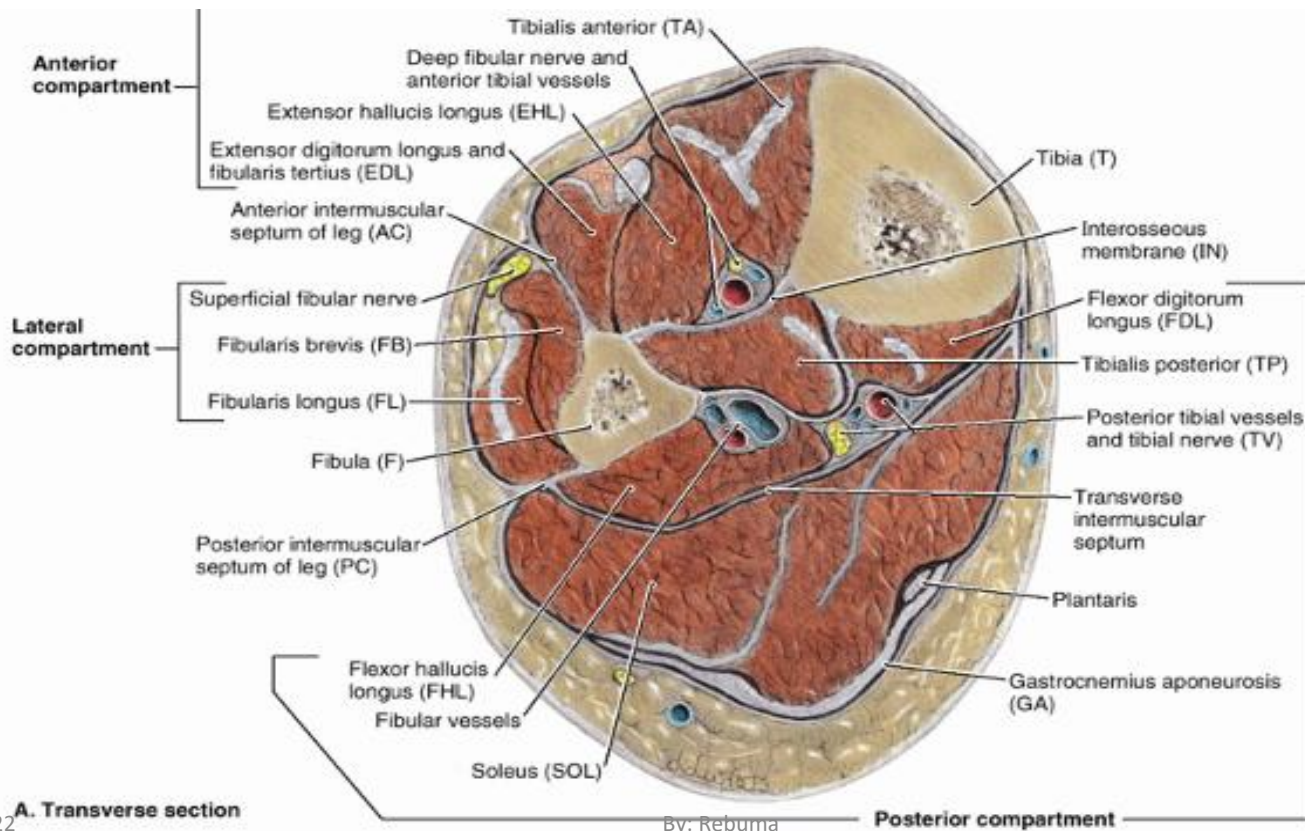
Neurovasculature in the lat. Comp.

- ❖ The **superficial fibular** nerve, the nerve in the lateral compartment, is a terminal branch of the common fibular nerve
- ❖ The lateral compartment of the leg does not have an artery coursing through it



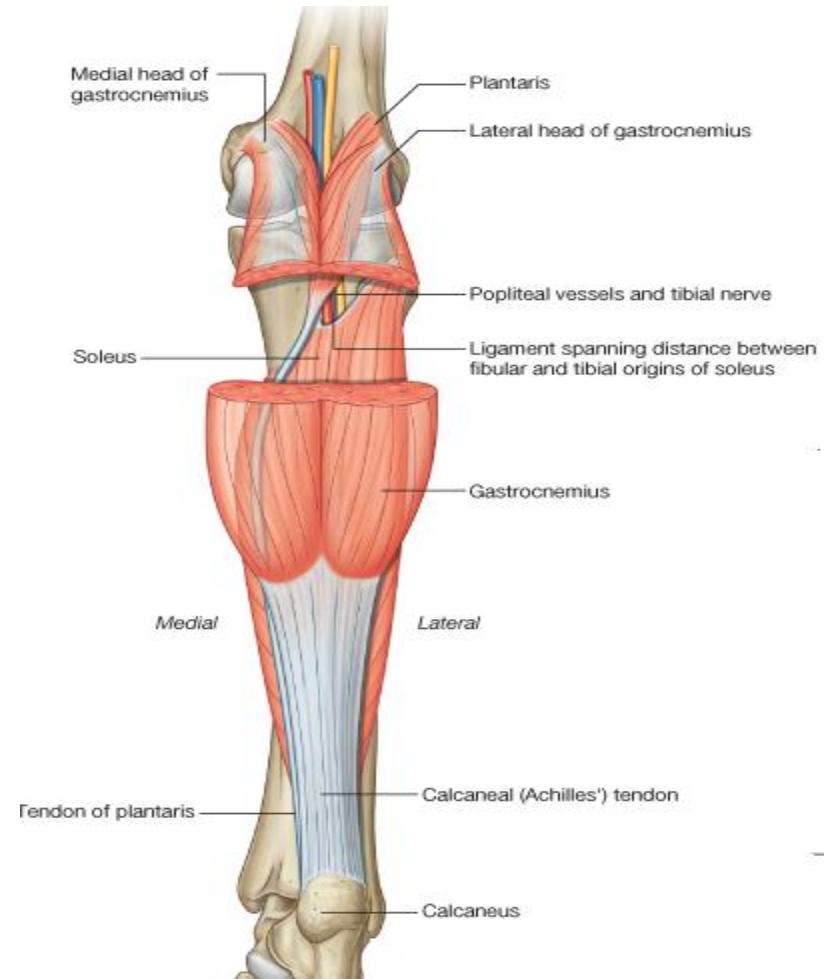
Posterior Compartment of Leg

- Is **Plantar flexor** compartment
- is the largest of the three leg compartments
- divided into **superficial** and **deep** sub compartments/muscle groups



The superficial group

- Includes the **gastrocnemius**, **Soleus**, and **plantaris**, and forms a **powerful muscular mass in the calf**.
- The two-headed of gastrocnemius and the Soleus share a common tendon, **the calcaneal tendon**



Gastrocnemius

- Origin: has 2 attachments just superior to the lateral and medial epicondyles of the femur
- Insertion
 - Ends in the calcaneus tendon (Achilles' tendon) which it shares with the soleus and plantaris
 - These 3 muscles form triceps surae
 - Tendo calcaneus inserts into the middle of the posterior surface of the calcaneus
- **Action** - plantar flexes foot, flexes knee
- **Nerve supply**: tibial nerve

Soleus

- Origin - proximal part of the posterior surface of the fibula and the soleal line of the tibia and interosseus membrane
- Insertion - calcaneus via calcaneal tendon
- Action - **plantar flexes ankle**
- Nerve supply: **tibial nerve**

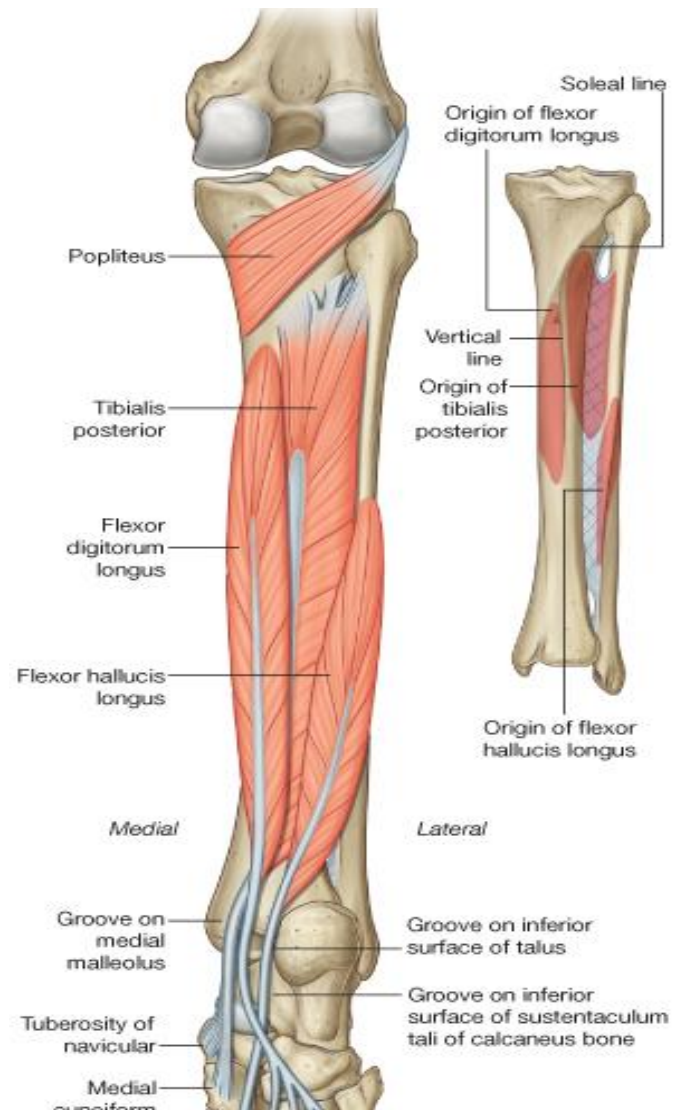
Plantaris

- lies between the gastrocnemius and the soleus
- Origin - posterior femur above lateral condyle; near the lateral head of the gastrocnemius
- Insertion - calcaneus via tendo calcaneus
- Action - **plantar flexion, knee flexion**
- Nerve supply: **tibial nerve**

THE DEEP GROUP

- Four muscles make up the deep group in the posterior compartment

1. Popliteus.
2. Flexor digitorum longus.
3. Flexor hallucis longus.
4. Tibialis posterior.



THE DEEP GROUP

Muscle	Proximal attachment	Distal attachment	Innervation	action
Popliteus	Lateral femoral condyle	Posterior surface of proximal tibia	Tibial nerve	Unlocks knee joint (laterally rotates femur on fixed tibia)
Flexor hallucis longus	Posterior surface of fibula	Base of distal phalanx of great toe	Tibial nerve	Flexes great toe, plantar flex ankle
Flexor digitorum longus	Medial side of posterior surface of the tibia	bases of distal phalanges of the lateral four toes	Tibial nerve	Flexes lateral four toes, plantar flex ankle
Tibialis posterior	Posterior surfaces of interosseous membrane and adjacent regions of tibia and fibula	Primarily Navicular and also on cuneiform, Cuboid, base of 2-4 metatarsal	Tibial nerve	Inversion and plantarflexion of foot

Blood vessel and nerves

✓ **Posterior tibial artery**

- It supplies the posterior and lateral compartments of the leg
- Ends by dividing into medial and lateral plantar arteries

➤ **Veins**

- Deep veins in the posterior compartment generally follow the arteries.

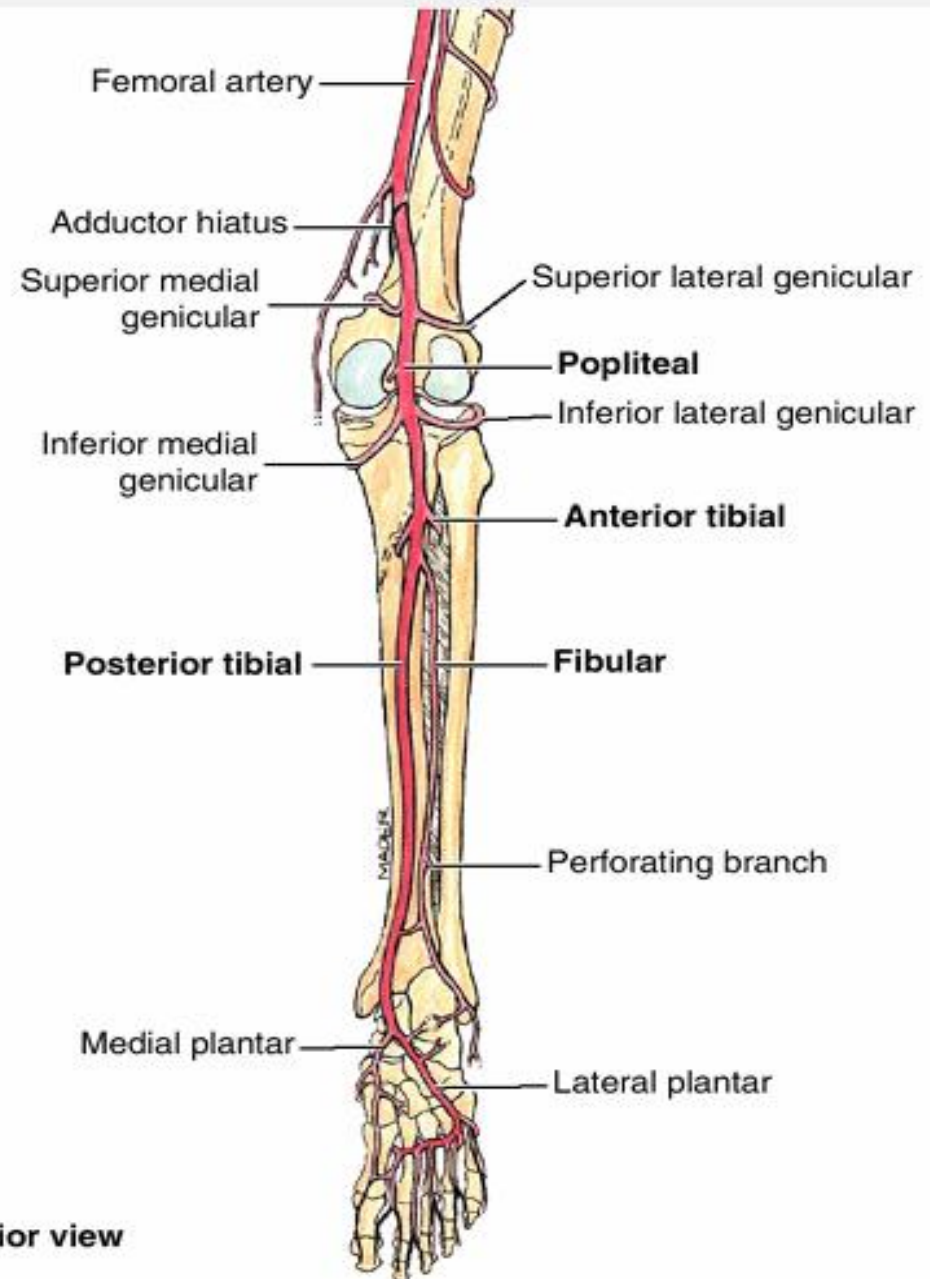
Neurovasculature in the pot. Comp

✓ **Tibial nerve**

- The nerve associated with the posterior compartment of leg is the *tibial nerve*.

Arteries of the leg

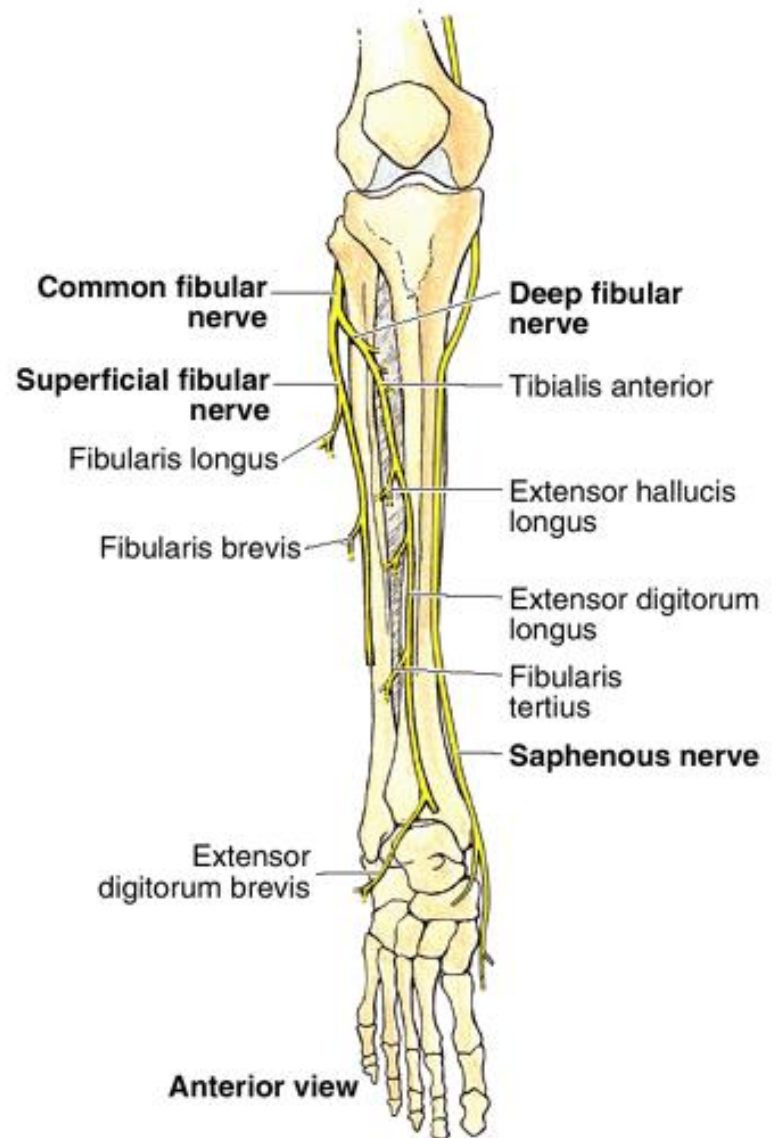
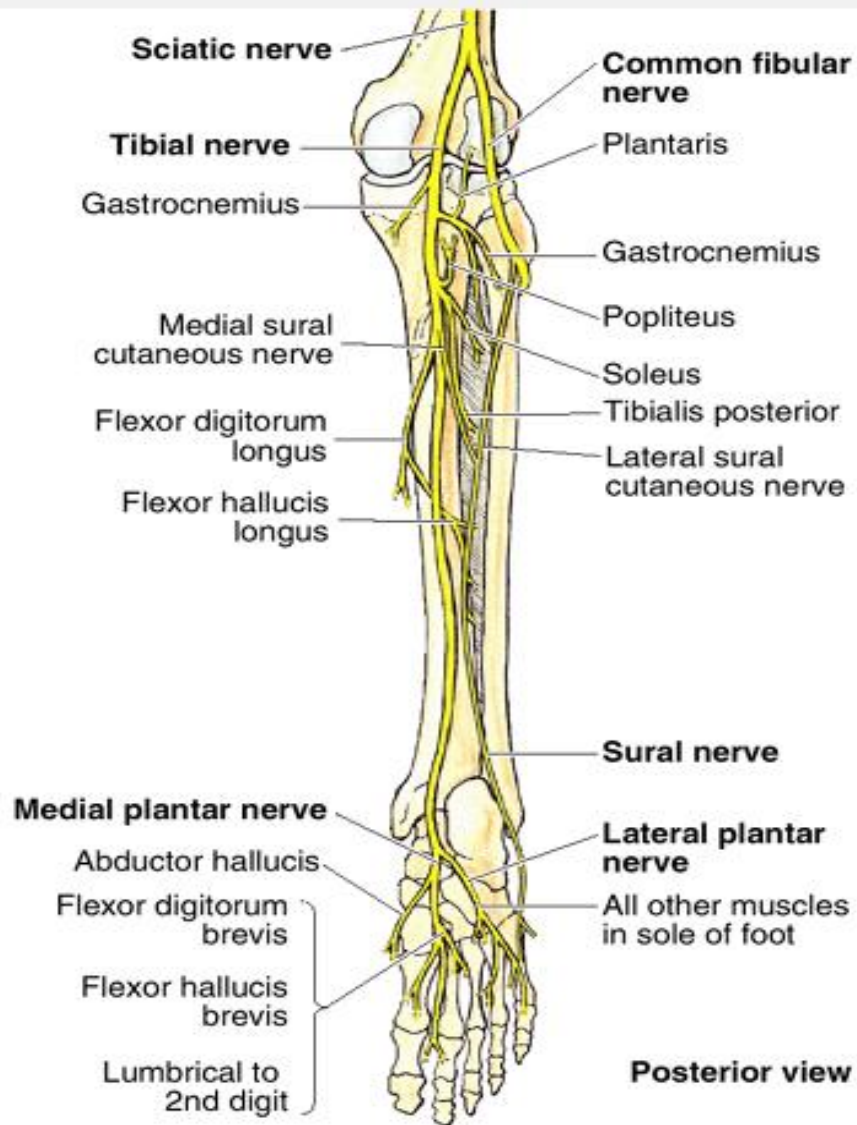
- Popliteal
- Anterior tibial
- Posterior tibial
- Peroneal



Posterior view

Nerves of the leg

- **Saphenous**:- skin on medial leg and foot
- **Sural**:-from tibial and common fibular nerve; skin on posterior and lateral leg and lateral foot
- **Tibial**:- arise from sciatic nerve; Posterior compartment muscles of the leg and knee joint;
- **Common fibular**:- from sciatic nerve; skin on lateral leg and knee joint
- **Superficial fibular**:- fibularis longus, brevis and skin on distal lateral leg and dorsum of foot
- **Deep fibular**:- anterior muscles of leg, dorsum of foot and skin of first interdigital cleft and joints

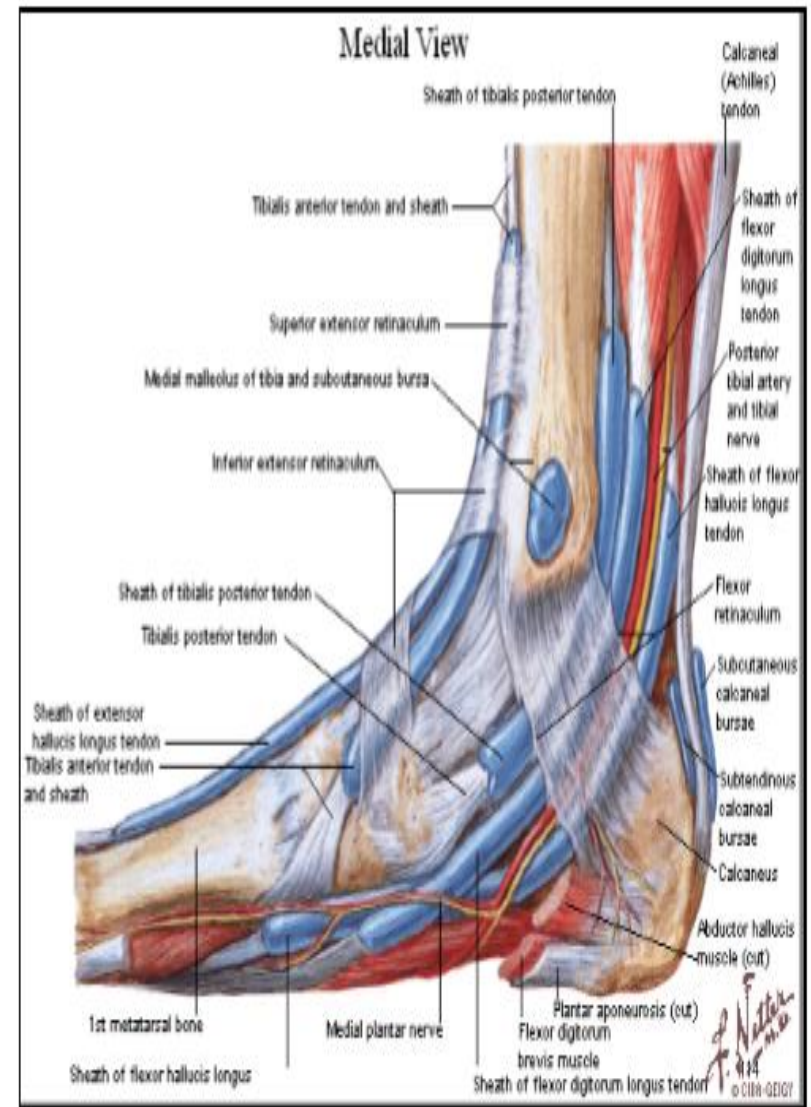
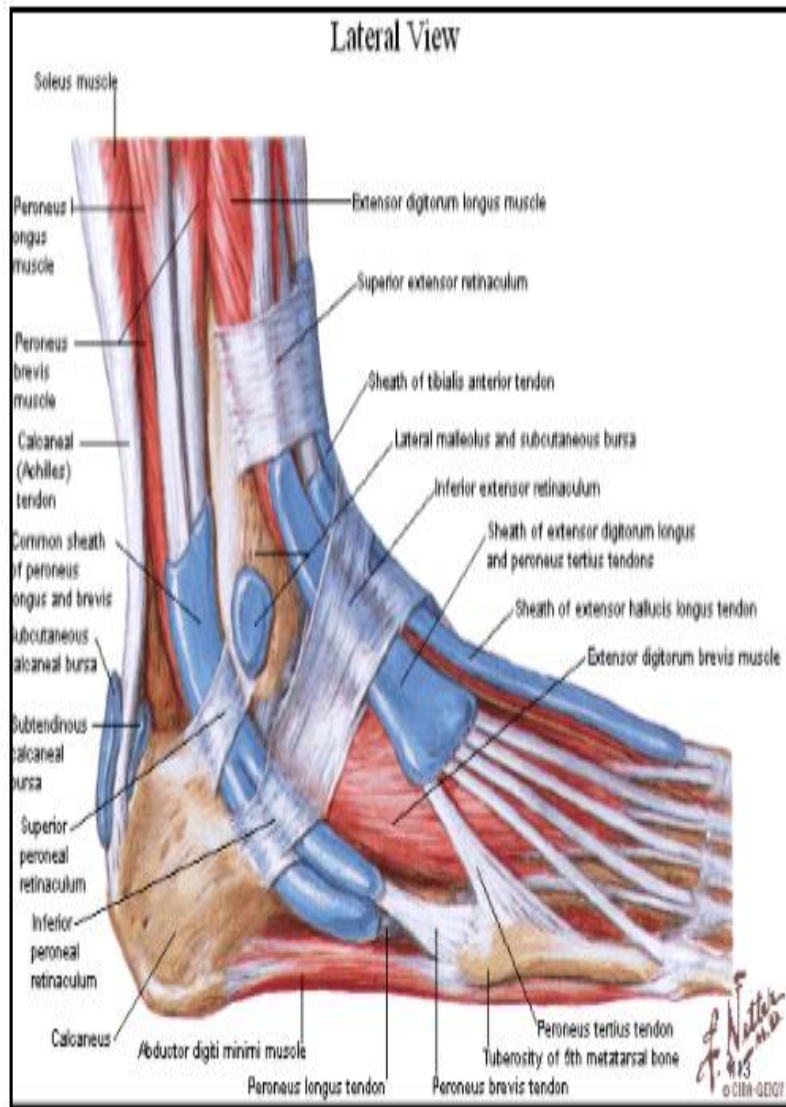


Injury to common fibular nerve

- The common fibular nerve is the most commonly injured nerve in the lower limb, mainly because it winds superficially around the neck of the fibula, leaving it vulnerable to **direct trauma**
- This nerve may also be severed during **fracture of the fibular neck** or severely stretched when the **knee joint is injured** or dislocated
- Severance of the common fibular nerve results in paralysis of all **muscles in the anterior and lateral compartments of the leg**
- The loss of dorsiflexion of the ankle causes **foot drop**, which is further exacerbated by unopposed inversion of the foot
- The toes do not clear the ground during the swing phase of walking

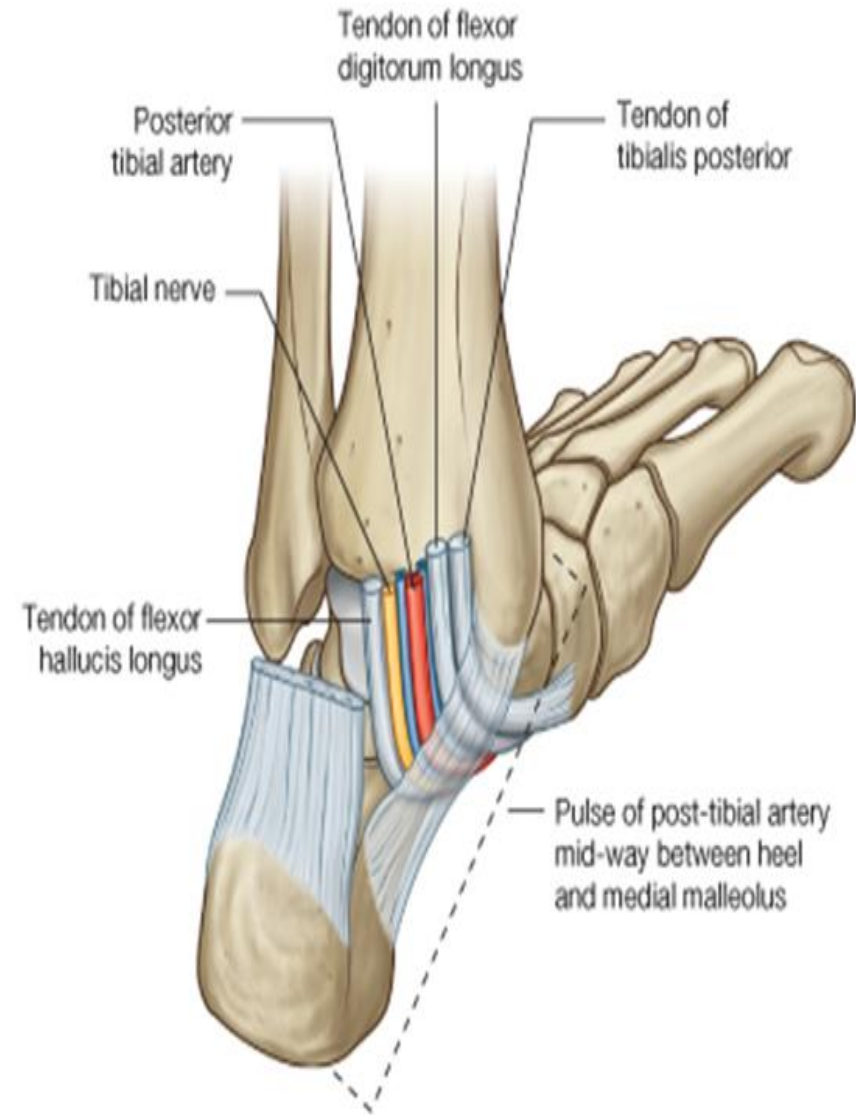
THE LEG

- At distal end, the fascia crura thickens to form retinacula to hold extensor, flexor and fibular muscles of the leg
- **Extensor retinaculum:** two:- **superior & inferior**
 - Superior: lie b/n distal part of tibia & fibula
 - Inferior: Y- shaped, calcaneus to medial malleolus (upper) & deep fascia (lower)
- **Peroneal retinaculum:** two: superior and inferior extend from the lateral malleolus to the lateral side of calcaneus
- **Flexor retinaculum:** b/n medial malleolus and medial side of calcaneus



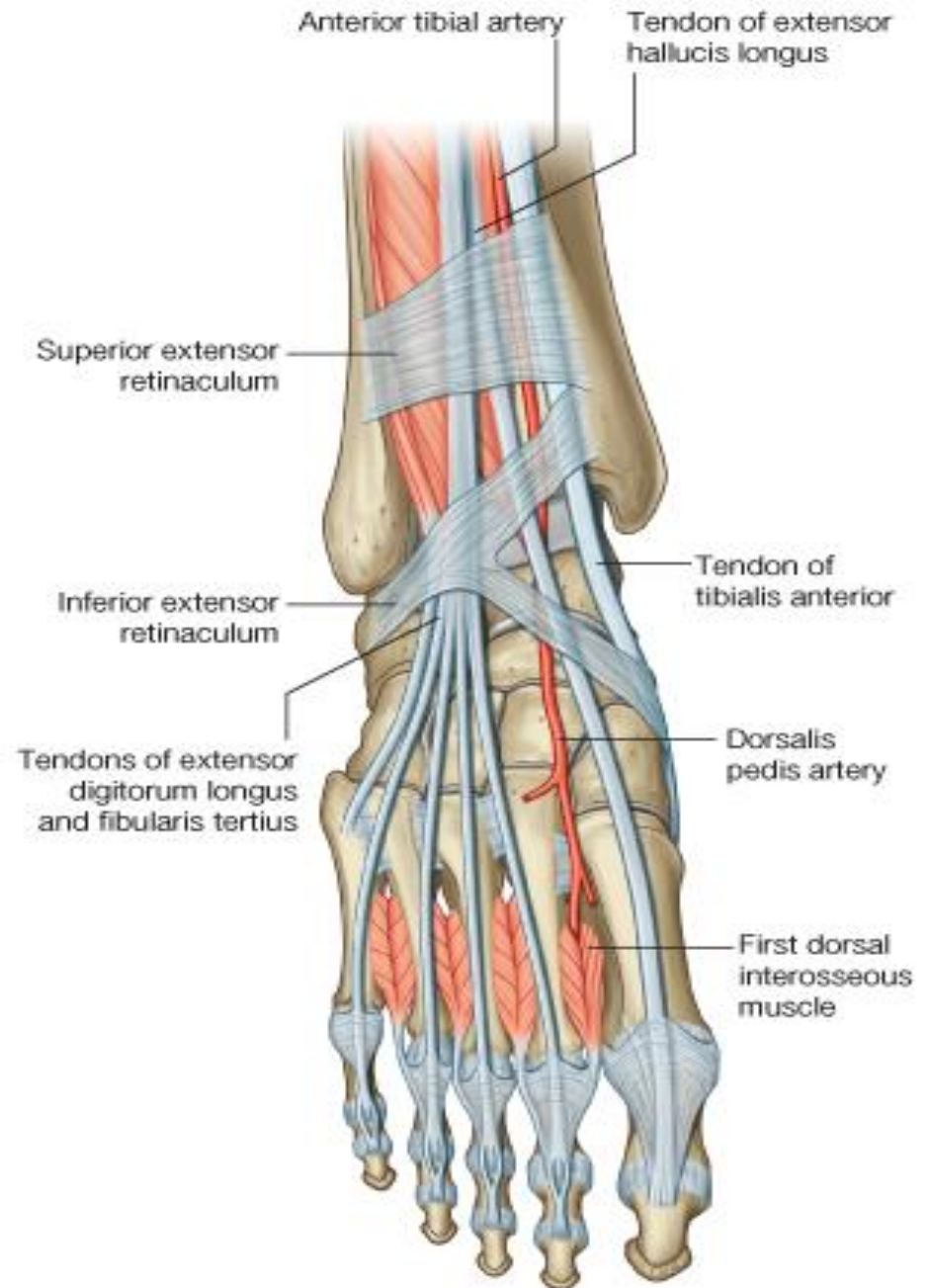
Flexor retinaculum

- From Medial malleolus to calcaneus
- Content – from medial to lateral
 - Tendon of tibialis posterior
 - Tendons of flexor digitorum longus
 - Termination of posterior tibial artery and beginning of lateral and medial plantar arteries
 - Termination of posterior tibial nerve and beginning of medial and lateral plantar nerves
 - Tendon of flexor hallucis longus



Extensor retinaculum

- Has superior and inferior portions
- Prevents the tendons of peroneus tertius, extensor hallucis longus and extensor digitorum longus from bowstringing during contraction





Thank u!!